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Uncovering Hierarchical Structure in Long-Term Natural Rodent Behaviour

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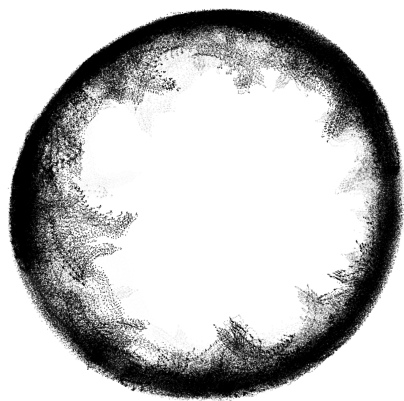
To my mother, for every mountain she has climbed, and those still ahead.

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I am sincerely grateful to Professor Mathis for the guidance and feedback throughout this project, and for the opportunity to work at the forefront of machine learning applied to behavioural analysis. I also thank the entire Mathis Group for the enriching and welcoming research environment I have been part of these past months, and owe particular thanks to Michal, Arash and Andy for their very much appreciated guidance.

We shall not cease from exploration
And the end of all our exploring
Will be to arrive where we started
And know the place for the first time.

—T. S. Eliot, *Little Gidding*, *Four Quartets* (1942)



Superposition of 27 2D keypoints across 29,915 frames (approximately 20 minutes) for subject HDP-002646 in old age, recorded during the habituation phase as the animal explored the circular arena.

Abstract

Natural, unconstrained behaviour is among the most complex expressions of an individual’s internal state, simultaneously reflecting an organism’s sensorimotor, motivational, and cognitive condition. This complexity is often described in terms of hierarchical organisation, yet it is rarely recovered in full. In fact, while recent advances in pose estimation and segmentation allow some aspects of it to be retrieved, the structure recovered is often pre-imposed by the model applied to the behaviour of interest. In this thesis we introduce two extensions to hBehaveMAE, a hierarchical masked autoencoder for unsupervised behavioural representation learning. On the synthetic benchmark Shot7M2, a nested dropout strategy is shown to recover hierarchical organisation without any prior spatio-temporal structure. On a multi-strain, multi-age, multi-phase open-field arena dataset, a length-agnostic implementation, using YaRN positional embeddings, enables variable-length inference over long, naturalistic recordings and yields behavioural embeddings that strongly discriminate between 80 different strains, while also capturing age- and phase-related structure. We additionally discuss the spatio-temporal definition of hierarchy, relating the strength of the hierarchical signal present in the data to how strongly it correlates with the hierarchy imposed by the model, and characterise the nested structure observed in the behavioural space extracted from an open-field arena dataset, advocating for less structural bias when studying the complexity of natural behaviour.

Keywords: Behaviour Analysis, Unsupervised Learning, Pose Estimation, Masked Autoencoders, Hierarchical Representations

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Chapter 1

Introduction

Natural, unconstrained behaviour is among the most complex expressions of an individual’s internal state, simultaneously reflecting an organism’s sensorimotor, motivational, and cognitive condition. This makes it a crucial readout for biomedical research, from tracking ageing processes and strain differences to evaluating disease models and drug responses. It is widely described as hierarchically organised, built from atomic motor primitives that combine into actions, and actions into longer behavioural episodes, across nested spatial and temporal scales. Yet when studied, the complexity of this hierarchy is rarely fully recovered.

Behaviour is traditionally studied through manual observation and hand-crafted ethograms which, while benefiting from expert knowledge, are limited by observer bias and the narrow scope of their pre-defined categories. More recently, by simplifying behaviour to pose estimation, it can also be studied in a bottom-up fashion, extracting structure from the data itself in a less biased way. This unsupervised approach is interesting precisely because the most informative behavioural differences may be ones that have not yet been conceptualised. In practice, however, the structure these methods recover is often still pre-imposed by the model applied to the behaviour of interest, whether through the number of behavioural levels assumed, their spatio-temporal resolution, or the choice to discretise a fundamentally continuous signal [26]. The challenge is therefore to let behavioural structure emerge with as little prior structure as possible.

hBehaveMAE [34], a hierarchical masked autoencoder for unsupervised behavioural representation learning, moves in this direction by learning a continuous representation directly from pose trajectories, recovering behavioural structure from a reconstruction objective rather than from a predefined segmentation. hBehaveMAE, however, still imposes pre-assumed structure: first temporally, by fixing the length of the studied behaviour, but more importantly through fixed spatio-temporal pooling kernels that dictate the resolutions of its hierarchical layers.

This thesis builds on hBehaveMAE to ask how far an unsupervised, reconstruction-based model can study behaviour with as few structural assumptions as possible, addressing each of these assumptions with a dedicated extension. A nested dropout strategy, validated on the synthetic benchmark Shot7M2, recovers hierarchical structure without relying on the model’s spatio-temporal pooling kernels. A length-agnostic implementation, using YaRN positional embeddings, lifts the fixed-length constraint and yields an improved embedding space. Applied to a multi-strain, multi-age, multi-phase open-field arena dataset, it strongly discriminates between 80 strains while also capturing age- and phase-related structure. Notably, the hierarchy that emerges from this natural behaviour is not the spatio-temporal one the architecture was designed to capture, but an abstraction

hierarchy, separating coarse factors such as strain and age from finer, experiment-related differences. This emergent organisation motivates the broader argument of this thesis: that hierarchy should be allowed to emerge from the data rather than be fixed a priori.

The thesis is organised as follows. Chapter 2 reviews the hierarchical nature of behaviour, pose estimation, state-of-the-art unsupervised segmentation methods, and recent advances in masked autoencoders. Chapter 3 describes the Arena dataset and the Shot7M2 and MABe22 benchmarks. Chapter 4 details hBehaveMAE, the proposed extensions, and the evaluation strategies. Chapter 5 presents the experimental results, and Chapter 6 discusses their implications.

Chapter 2

Background & Related Work

2.1 The Hierarchical Nature of Behaviour

Behaviour was defined by Levitis, Lidicker Jr, and Freund [20], through a comprehensive survey of the field, as

“the internally coordinated responses (actions or inactions) of whole living organisms to internal and/or external stimuli.”

On this basis, many have tried to describe this coordinated response as a hierarchical structure. Some discuss the temporal organisation of behaviour across multiple nested scales: from sub-second motor primitives [39], to action sequences [23], to behavioural episodes [1], circadian rhythms, and sometimes even lifelong ageing trajectories [4]. Hierarchical behaviour is also described spatially, where complex movements involving the coordinated articulation of multiple limbs and joints can themselves be decomposed into joint-specific sub-movements [34]. Curriculum learning, where complex tasks are naturally decomposed into sub-tasks as part of the learning process, is a good example of this spatial hierarchical organisation [9]. This spatio-temporal, multi-scale structure is not simply a convenient modelling choice: it is increasingly viewed as a fundamental property of motor behaviour that any quantitative behavioural analysis must account for [17, 23].

As a first construction block, the concept of behavioural *syllables* or *movemes*, the atomic units of movement from which more complex actions are composed, has become central to quantitative behavioural analysis [23]. Although their temporal resolution is relatively agreed upon, ranging from 300 to 720 ms in rodents [23, 40, 39, 26], the number of these construction blocks and their actual discrete nature are debated [39, 26]. In addition, no consensus exists regarding the higher levels of this hierarchy. The number of relevant hierarchical layers and their associated spatio-temporal resolution vary substantially across studies. Huang et al. [17] define three levels: poses (~ 0.17 s), movements (~ 0.8 s), and ethograms (~ 3 s), validating the biological relevance of each scale through 3D motion capture. Stoffl et al. [34] define three levels: movemes, actions, and activities, spanning timescales from a few frames (~ 100 ms) to entire episodes of 1800 frames (60 s). Weinreb et al. [40] extend the atomic syllable concept by defining behavioural states as higher-order Markovian modes governing the expression of these syllables, effectively partitioning spontaneous behaviour into two distinct hierarchical layers: the sub-second syllable and the multi-second behavioural state. They additionally show that within one-hour behavioural recordings, these two levels of hierarchy are sufficient to describe the organisation of spontaneous behaviour.

This two-level hierarchical organisation of behaviour, separating behavioural states from atomic actions, has a counterpart in the brain. The dorsomedial prefrontal cortex (dmPFC) has been shown to encode behavioural state related information, while the dorsolateral striatum (DLS) has been shown to correlate with individual syllables [40, 23]. Notably, lesions of the dmPFC have been shown to produce no visible changes in the raw kinematic form of individual actions, but instead affect the duration and usage of syllables within behavioural states, as well as reduce the frequency of less common behavioural states such as grooming and local investigation [40]. This dissociation suggests a possible separate control at different hierarchical levels, where the action signal may be preserved in one region while its selection and organisation into higher-order states is governed by another. This view is in part discussed in the computational model of Holroyd and McClure [15], who define a three-level hierarchy of actions, options, and meta-options, assigning action selection to the striatum, option selection to the anterior cingulate cortex, and meta-option selection to the prelimbic cortex. They anchor this hierarchy in the reinforcement learning framework, where each level integrates reward signals at a different temporal resolution. This connects back to the definition of Levitis et al. defining behaviour as a response to a stimulus. In fact, a response to a stimulus implies a decision process, and the complexity of that decision, from a simple reflex to a multi-step goal-directed plan, determines how many hierarchical levels may be required. Depending on the goal, a more or less complex behaviour is initiated, itself decomposed into subtasks at multiple temporal and joint resolutions [15, 40].

This hierarchical organisation of the brain is, however, actively debated. Han et al. [13] demonstrate that brain regions previously thought to encode only low-level kinematic information are in fact sufficient to generate hierarchically structured behaviour, with the hierarchy emerging from the dimensional organisation of local cortical dynamics rather than from any anatomical hierarchy between brain regions. In their framework, low-dimensional neural dynamics encode high-level behavioural composites, while higher-dimensional dynamics encode fine-grained kinematics. This non-hierarchical view of the brain is often referred to as the *shallow brain hypothesis*. This result challenges the assumption, common in both neuroscience and computational modelling, that hierarchical behaviour necessarily requires hierarchically organised neural substrates [13].

This lack of consensus, both about the neural basis of behavioural hierarchy and about the appropriate spatio-temporal resolution at which to describe it, makes the task of learning and understanding the different components of behavioural organisation exceptionally challenging. Resolving it requires behavioural data that is itself as unconstrained as possible: behaviour studied in controlled environments, shaped by experimental design, may impose the very structure one is trying to discover. Rising interest in natural behaviour reflects precisely this observation: free behaviour provides a richer and less biased readout of an individual’s state. [12, 3, 4]. In this regard, many studies have attempted to study this hierarchy in a bottom-up, data-driven way, allowing hierarchical structure to emerge from the data itself without prior assumptions about the number of relevant levels or their spatio-temporal resolution [26, 41, 34]. This thesis aims to improve upon this approach by reducing the need for structural assumptions about behaviour.

2.2 Pose Estimation as the Input Modality

Behavioural studies were for a long time centred around manual observation and hand-crafted ethograms. These ethograms, although profiting from the knowledge of experts, are not only time-consuming but also inherently limited by observer bias, the narrow

DeepLabCut: markerless tracking toolbox

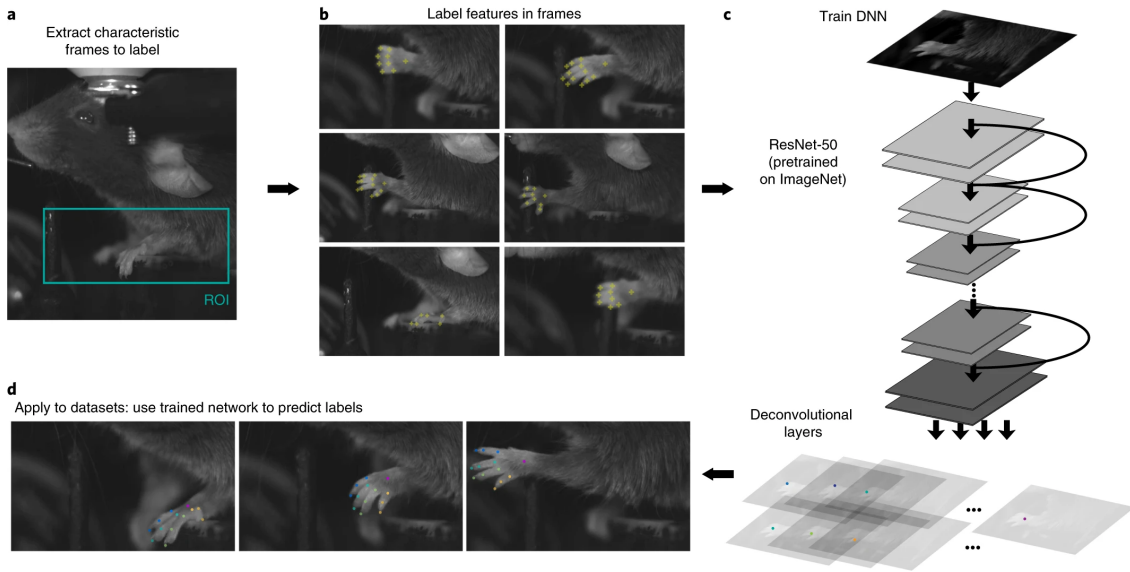


Figure 2.1: (a) Characteristic frames are extracted from the recorded video, and a region of interest (ROI, teal box) is defined around the body part to be tracked. (b) User-defined features — here the digits and joints of the rodent’s hand — are manually labelled across a small set of representative frames. (c) A deep neural network, comprising a ResNet-50 backbone pretrained on ImageNet followed by deconvolutional layers, is trained on the labelled frames to predict a spatial probability map (heatmap) for the location of each body part. (d) The trained network is then applied to new datasets to automatically predict the positions of the labelled body parts. Reproduced from Mathis et al. (2018), *Nature Neuroscience* 21:1281–1289, with permission from Springer Nature (licence no. 6281231443138) [24].

scope of pre-defined behavioural categories, and the practical impossibility of capturing fine-grained or rapid movements by eye alone [12]. Recent advances in machine learning and computer vision have given rise to a new data modality that has now become standard in the field of quantitative behaviour analysis: markerless pose estimation. Rather than processing raw video directly, pose estimation reduces the visual signal to a sparse set of anatomically defined spatial landmarks. These landmarks, often called keypoints, provide a compact and interpretable representation of body configuration at each frame.

The spatial and temporal resolution of these keypoints is specific to the experimental question. In rodents, for example, precise tracking of a single paw is sufficient for a controlled lever-press reward experiment [24] (Fig. 2.1), while a full-body skeleton of 8 to 27 keypoints is more appropriate for capturing the diversity of spontaneous, naturalistic behaviour [39]. Keypoints can be estimated in 2D from a single overhead camera, or in 3D from multi-camera setups, with the choice depending on the granularity of the behaviours of interest [17].

DeepLabCut, developed by Mathis et al., has become the standard tool for markerless pose estimation in neuroscience [24]. It produces reliable keypoint tracks from as few as a few hundred human-labelled frames by fine-tuning deep convolutional networks pre-trained on ImageNet, building on the ResNet backbone. Other tools such as SLEAP offer similar capabilities with different annotation workflows and multi-animal tracking support [28], but the underlying principle: supervised learning from sparse human labels, is shared across methods.

By reducing the complexity of the visual signal to a small set of coordinates, pose estimation inevitably discards potentially relevant information. It should therefore always be interpreted as a useful simplification of behaviour rather than a complete description.

That said, this simplification brings two important advantages: it makes behavioural data far more compact and computationally tractable, and it produces representations that are more objective and reproducible than manual ethograms, since the same keypoint tracker applied to different recordings yields directly comparable outputs.

Once keypoints are obtained, several preprocessing choices are typically applied before behavioural analysis. Raw coordinates can be studied in arena space, preserving positional information within the experimental environment [3, 37]. However, many methods favour egocentric alignment, rotating each frame so that a reference body axis (typically the neck-to-tail-base vector) is aligned with a fixed direction and subtracting the body centre position, in order to disentangle body posture from arena position and heading direction, and to facilitate comparison across individuals [39, 6, 26]. Notably, this disentanglement allowed Huang et al. [17] to study what they define as locomotor and non-locomotor movement separately, finding that locomotion accounted for only 20.6% of studied behaviour.

2.3 Unsupervised Behavioural Segmentation

2.3.1 The Segmentation Problem

As discussed previously, although continuous, behaviour is often regarded as built from discrete blocks of actions and decisions. Many approaches have been developed to retrieve these behavioural blocks by segmenting the pose estimation. Starting from human labelled ethograms, in the form of structured catalogues of predefined behavioural categories such as locomotion, rearing, grooming, and freezing [42] in rodents [10]. These ethograms could then be used to compare differences in behavioural usage across, for example, species, age groups, or experimental paradigms [7, 25, 10]. The already listed limitations of this approach, combined with recent advances in machine learning, has motivated a new approach: unsupervised learning. By simplifying the behavioural data modality to pose estimation, behaviour can be studied in a bottom-up fashion, extracting an ethogram from the data itself in a less biased way.

This segmentation, although now widely used, is also criticised by many for its artificial nature. Although the recovered units have been shown to correlate with neural processes [23, 40], these segmentations have also been shown to impose relatively arbitrary boundaries on a continuous behavioural space [39]. Mlost et al. [26] show that most segmentation methods yield very low and sometimes even negative Silhouette scores, a metric reflecting how representative of the underlying space the resulting clusters are [26]. For instance, extracting syllables, one might recover slow left turning versus fast left turning: it is debatable whether these constitute genuinely distinct behavioural units rather than two points along a continuum of the same action. In fact, although the temporal grounding of these units in neural processes provides some biological validation, the number of recovered blocks varies substantially across studies, ranging from as few as 15 to over 100 in others [26, 39]. Despite this artificial character, behavioural segmentation remains highly useful as a tool for further simplifying pose estimation for downstream analysis, and as a common reference format that enables comparison across methods and studies [26].

2.3.2 Single-Timescale Methods

Different approaches exist to segment pose estimation data, but all rely on a form of clustering. Methods differ in the data that is clustered and the clustering algorithm applied. Both aspects are crucial in defining the resulting behavioural units: the choice of

data representation defines the behavioural space, while the clustering method defines the boundaries and distance metric used to segment it. Methods such as B-SOiD [16] compute simple kinematic features directly from keypoints, specifically, 15 spatiotemporal displacement and angular change measures alongside six inter-keypoint distances, which are then reduced through UMAP to obtain a lower-dimensional manifold in which they are clustered with HDBSCAN [16]. Some approaches apply simple PCA or no dimensionality reduction at all, as in Behavioural Factor Analysis (BFA) [38], allowing for a more interpretable analysis of cluster distributions.

Another common approach, VAME [22], also transforms the pose estimation non-linearly, but through a variational autoencoder. This autoencoder learns to encode behaviour in a bottle-necked latent space by reconstructing behaviour from it. The resulting latent space is then segmented by a hidden Markov model. Finally, the approach that has shown the strongest performance and has become the most widely used is Keypoint-MoSeq [39]. This method applies PCA to the pose estimation and clusters the resulting representation through an autoregressive hidden Markov model. Unlike the methods above, Keypoint-MoSeq uses a generative model that explicitly separates keypoint noise from behavioural signal, identifying syllable boundaries at natural discontinuities in pose dynamics rather than at arbitrary cluster frontiers. This property, combined with strong alignment shown between the recovered syllables and neural activity in the DLS [23], makes Keypoint-MoSeq the current state of the art for unsupervised behavioural segmentation. A systematic comparison of B-SOiD, BFA, VAME, and Keypoint-MoSeq by Mlost et al. [26] confirms this ordering.

Although all these methods have shown promising results, they share a fundamental limitation: they segment behaviour at a single timescale, focusing exclusively on the atomic unit of movement. The recovered syllables typically include locomotion, turning, or grooming bouts, with no representation of the higher-order structure in which these units are organised [40, 34, 1].

2.3.3 Multi-Timescale Approaches

Some methods have been implemented to try and segment continuous pose estimation into hierarchical blocks. BAMS [1], for example, defines two different timescales of interest: a short-term scale at 31 (1s. on the MABe22 dataset) frames and a long-term scale at around 683 frames (23s.). These two representations are obtained by running the pose estimation through two separate Temporal Convolutional Networks. The training process works similarly to VAME [22] in that it uses a reconstruction objective, but BAMS introduces a key novelty: rather than predicting the exact pose, it predicts what the authors call the *histogram of actions*, where actions represent kinematic quantities such as the linear velocity of a joint or the angular velocity of the head. The model therefore learns the future distribution of these actions within a given temporal window. This leads to a qualitative difference in the encoded representations: whereas VAME latents are trained to predict the pose sequence with a focus on the temporal ordering of actions, the latents obtained from BAMS constitute a more abstract representation of the behavioural state as a distribution over actions, with no dependence on their temporal or sequential organisation. This method held the best results on the MABe22 challenge [1, 36] for some time, where representation quality is evaluated through linear probing, training a classifier to predict either sequence-level or frame-level labels from frozen embeddings. Interestingly, prediction was strongest when both short- and long-term representations were concatenated. However, the expected pattern: short-term representations being

more informative for frame-level labels and long-term representations for sequence-level labels, was not consistently observed. In fact, the long-term layer sometimes outperformed the short-term one on frame-level prediction.

Another approach is TS2Vec [43], which builds multi-scale representations through contrastive learning rather than a reconstruction objective. Also leveraging Temporal Convolutional Networks, it extracts n representations of a behavioural time series by iteratively applying a max-pooling kernel of size 2 along the temporal axis n times, producing a hierarchy of progressively coarser representations. Two complementary contrastive losses are defined: a *temporal contrastive loss*, which takes representations at the same timestamp from two augmented views of the same time series as positives and representations at different timestamps as negatives; and an *instance-wise contrastive loss*, which treats representations from different time series at the same timestamp as negatives. These losses are accumulated across all pooling levels and averaged, simultaneously training all representational scales in a single pass [43]. While BAMS enforces specific temporal resolutions for the different levels of the behavioural hierarchy, fixed by the convolutional kernel sizes and its two-level definition of short- and long-term scales, TS2Vec learns to build representations at all temporal scales equally without any fixed prior assumptions on the number or duration of relevant levels.

Finally, building upon the widely used MoSeq method, shMoSeq (state-based hierarchical MoSeq) [40] defines a second hierarchical level above the syllable. This state arises from the natural observation that the Markovian transition probabilities between syllables quickly reach chance level, suggesting that syllable sequences alone are insufficient to capture the full structure of behaviour. To explain this observation, a behavioural state is introduced as a higher hierarchical layer that defines the probability distribution over syllables in the form of a transition matrix. As an improvement over standard hierarchical hidden Markov models (HHMMs), shMoSeq decomposes this transition matrix into two components: one representing the baseline transition probabilities of a hypothesised base behaviour, and one specific to each behavioural state, encoding the deviation from this baseline. This model has been shown to outperform HHMMs when a large number of syllables are observed [40]. This method is particularly interesting both for its connection to neural theories separating action probability encoding in the prefrontal cortex from action selection in the striatum [15, 23], and for its bottom-up construction: atomic units of behaviour are first recovered, their temporal organisation is then observed, and higher-level structure is then defined from this foundation. This approach could in principle be extended to any number of hierarchical layers, allowing the full behavioural hierarchy to be reconstructed in a bottom-up fashion.

On the other hand, this method can also be criticised. First, it relies heavily on hyperparameters: at the syllable level, the stickiness parameter κ controls the typical duration of recovered syllables [39], an analogous parameter `trans_kappa` governs the typical duration of behavioural states, and `n.states` sets the number of states, none of which are inferred from the data [40]. Secondly, as a consequence of its bottom-up construction, it never models behaviour jointly across levels: the number of behavioural states does not influence the syllables found at the lower level, meaning the two levels are optimised independently rather than in a truly joint hierarchical model.

In conclusion, most hierarchical methods rely on preconceptions about the structure of behavioural hierarchy. This gap directly motivates the approach developed in this thesis.

2.4 Masked Autoencoders for Behavioural Representation

The methods discussed so far share a common limitation: they either impose a fixed representational structure on behaviour or rely on explicit segmentation as a prerequisite for hierarchical analysis. A complementary approach is to learn a continuous, general-purpose behavioural representation directly from pose trajectories, without committing to segmentation upfront. Reconstruction-based objectives, as used in VAME and BAMS, have shown that forcing a model to recover behavioural signals from a compressed representation produces latent spaces that are both compact and behaviourally meaningful. hBehaveMAE extends this principle by combining the reconstruction objective with masked autoencoding and transformer architectures, which have recently demonstrated strong representation learning across a wide range of sequential and structured data [21].

2.4.1 Masked Autoencoders

The masked autoencoder (MAE) pretraining paradigm, introduced by He et al. [14], defines a self-supervised objective in which a large fraction of input tokens are randomly masked and the model is trained to reconstruct the masked regions from the visible ones. The encoder processes only visible tokens, producing a compressed representation that must support plausible reconstruction of the missing content. The decoder, used only during training, reconstructs the masked tokens from the encoder output concatenated with learned mask tokens; at inference, only the encoder is retained.

The key driver of representation quality in MAE pretraining is reconstruction difficulty. At low masking ratios, adjacent visible tokens provide sufficient context for simple interpolation, and the encoder doesn't need to learn any compact or structured representation. At high masking ratios, typically 75% for images [14], local context is insufficient, and the encoder is forced to develop representations that capture global structure in order to support reconstruction. This difficulty-driven representation learning property shapes all subsequent work discussed in this section.

2.4.2 Hiera

Hiera [32] is a hierarchical vision transformer designed to combine the representational benefits of hierarchical pooling with the efficiency of MAE pretraining. Earlier hierarchical transformers required convolutional components and explicit spatial downsampling modules that were incompatible with the sparse token processing of MAE. Hiera removes these components entirely, applying only query (Q) pooling at stage transitions to progressively reduce the number of tokens across encoder stages.

Attention in Hiera is organised around the mask unit: the coarsest token granularity produced after full Q-pooling. Masking is applied at the mask unit level, and attention in early stages is restricted to within mask unit boundaries, ensuring that all tokens participating in attention are visible. This prevents attention from leaking into masked regions, a failure mode of window attention in sparse MAE pretraining, while also accelerating training. The pooling cascade produces a natural hierarchy of representations as a by-product: early stages encode fine-grained local features at high resolution, while later stages encode progressively more global and abstract features at coarser resolution.

2.4.3 hBehaveMAE as a Spatio-Temporal Hierarchical Model

hBehaveMAE is a masked autoencoder introduced by Stoffl et al. [34], built upon the Hiera architecture [32]. Several modifications were introduced to adapt the model to multi-dimensional behavioural time series. First, the model was extended to support non-square pooling kernels, enabling temporal as well as spatial pooling. For example, the first-stage kernel for Shot7M2 is $(2, 1, 4)$, pooling over the temporal, individual, and keypoint dimensions respectively. Second, a multi-scale decoding strategy was introduced. Whereas the original Hiera reconstructs only from the final encoder stage, hBehaveMAE optionally employs a fusion head: a learned spatio-temporal pooling module applied to each intermediate stage, projecting its output to match the dimensionality of the last stage before summing all contributions and passing the result to the decoder.

Applied to hierarchical behaviour analysis, this architecture outperformed state-of-the-art methods on both MABe22 and hBABEL [36]. Results on the synthetic hierarchical behaviour dataset Shot7M2 further showed that different encoder levels specialise in distinct behavioural granularities: finer levels capture movemes, while coarser levels encode actions and activities. The promising results and the as-yet-untapped potential of the multi-scale decoding strategy motivate the use of this method in the present thesis.

2.5 Nested Dropout and FlexTok

Nested dropout, originally proposed by Rippel et al. [31], is a training procedure for imposing an ordered structure on otherwise unordered latent representations. Given a latent vector of dimension d , nested dropout randomly retains only the first k dimensions during each training step, where k is sampled from a distribution over $\{1, \dots, d\}$. Because the model must reconstruct its target from any prefix of the latent vector, it is forced to front-load the most globally important information into early dimensions and progressively refine it in later ones. The resulting representation is ordered in the sense that any prefix of length k provides the best k -dimensional approximation to the full representation, a property closely related to principal component analysis but learned end-to-end from a reconstruction objective.

FlexTok [2] demonstrates the power of this principle at scale for image tokenisation. Rather than compressing an image into a fixed-size token grid, FlexTok learns a 1D ordered token sequence of variable length: the first token captures the coarsest global structure of the image, and each subsequent token progressively refines the representation. This ordering is not hand-crafted but emerges entirely from applying nested dropout to the token sequence during training. The result is a flexible tokeniser: simple images can be reconstructed from a small number of tokens, while complex images require more. Crucially, the hierarchy induced by nested dropout is horizontal: it operates along the dimension of a single flat embedding vector. In FlexTok, there is no prior hierarchy between tokens; hierarchy emerges from the dropout alone.

2.6 Behavioural Phenotyping

The preceding sections have treated behavioural analysis primarily as a computational problem: how to segment, represent, and structure continuous behaviour. This section shifts to the biological question that motivates the application of these methods: what can behaviour, rigorously quantified, tell us about the internal state of an organism, and why does the quality of behavioural measurement matter for biomedical research?

2.6.1 Behaviour as a Phenotype

Behaviour is the most integrative phenotype available to the experimental biologist. Unlike molecular or imaging readouts that study a single biological layer, spontaneous locomotor behaviour in an open-field arena simultaneously reflects the organism's sensorimotor capacity, emotional state, motivational drive, and cognitive function, all without surgical intervention or external stimulation [10, 37]. This integrative sensitivity is both its greatest strength and its greatest challenge: behaviour is rich precisely because it is the downstream product of many interacting biological systems, making it difficult to attribute behavioural differences to specific mechanisms without careful experimental design and precise quantification.

The sensitivity of behaviour to genetic background has been established across decades of inbred strain research. Crawley et al. [10] documented extensive, reproducible differences across a panel of inbred mouse strains in tasks like locomotor activity, anxiety-related exploration, social behaviour, and learning. These differences are stable across laboratories and reflect the fixed genetic divergence between strains. At the other end of the temporal scale, Bedbrook et al. [4] recently demonstrated that behaviour is also sensitive to ageing trajectories: longitudinal behavioural recordings in *Nothobranchius furzeri* revealed that the full lifespan of an individual can be predicted from movement patterns recorded at a young age, and that distinct behavioural stages, separated by sharp transitions, mark the progression from maturity to late life [4, 7]. Together, these results establish that spontaneous behaviour in a simple open-field setting carries quantitative information about genetic background, developmental stage, and ageing, making it a powerful non-invasive window into organism state.

2.6.2 Inbred Mouse Strains as a Model System

Inbred mouse strains occupy a central role in preclinical biomedical research for a straightforward reason: their fixed genetic background enables reproducibility. Unlike outbred populations, where individual genetic variation introduces uncontrolled noise, mice of the same inbred strain are genetically near-identical, ensuring that differences observed between experimental groups are a result of the imposed intervention rather than uncontrolled genetic variance [10]. This property has made a small set of strains, most prominently C57BL/6J, the near-universal background for transgenic disease models.

Yet this narrow reliance on a handful of strains conceals a deeper problem. Strains differ substantially not only in molecular phenotypes but in their behavioural responses to disease, drugs, and environmental perturbations. A treatment effect demonstrated in C57BL/6J mice may not generalise to other genetic backgrounds, and phenotypes that are invisible in one strain may be easily detectable in another [10, 25]. This strain-by-treatment interaction is not only a technical nuisance: it is biologically meaningful, reflecting the reality that human patients also carry diverse genetic backgrounds that modulate treatment response.

A critical but often neglected prerequisite for any strain-comparative study is the establishment of reliable behavioural baselines. Understanding how each strain behaves under neutral conditions, what its spontaneous activity profile looks like, how it explores a novel environment, how its behaviour evolves with age, is a prerequisite for detecting any deviation from that baseline induced by a disease model, genetic mutation, or pharmacological intervention. Crawley et al. [10] recognised this explicitly, framing their strain survey not as an end in itself but as a foundation for interpreting future experimental results in the appropriate genetic context.

2.6.3 The Stylometry of Movement

The problem of characterising strain-specific behavioural style has a natural analogy in computational linguistics: stylometry, the identification of an author from the statistical patterns of their writing rather than its semantic content. In the behavioural domain, the question is similar: can the *style* of movement, the particular distribution of behavioural syllables, the characteristic way in which actions are sequenced and organised into higher-order states, uniquely identify a strain independently of where the animal happens to be positioned in the arena or what specific movement it is executing at any given moment [1]?

This framing has several practical consequences that connect directly to the methodological choices made in this thesis. In fact, in order to retrieve the stylometry of a specific behaviour, it is crucial to remove any non-behaviourally relevant information or bias. First, positional information should be separated from postural information: a strain difference in spatial occupancy reflects a preference for wall-following or centre avoidance, but a strain difference in the *form* of locomotion or the composition of grooming bouts reflects something deeper about the biomechanical and motivational organisation of movement. Egocentric alignment: removing the body centre position and orienting all frames to a fixed heading direction, disentangles these two sources of variation, allowing the downstream analysis to target behavioural style rather than spatial strategy. Second, body morphology differs systematically across strains in ways that are not behaviourally meaningful: size normalisation and keypoint subsampling are required to ensure that strain differences in the recovered behavioural representations reflect differences in movement dynamics rather than differences in physical scale.

2.6.4 The Case for Unbiased, Hierarchical Behavioural Analysis

The stylometric view of strain behaviour raises several immediate methodological questions: at which spatio-temporal resolution does this style emerge? Does the strain of an animal affect the broad distribution of behavioural states? Does it affect the distribution of actions within those states? Does it affect the actual execution of a given action? These questions are non-exhaustive and illustrate the range of complex effects that may be observed in natural behaviour; analogous questions apply to ageing processes, drugs, and experimental paradigms.

The classical answer, inherited from the ethogram tradition, has been to focus on a small number of defined behavioural categories: locomotion, rearing, grooming, freezing, and to measure the time animals spend in each [10]. This approach has produced a substantial body of comparative knowledge, but it suffers from a fundamental limitation: it studies stylometry at a single level. In addition, some behavioural categories retrieved from complex unsupervised methods disregard potentially relevant clusters by merging apparently similar or kinematically similar ones, potentially neglecting differences in the execution of a specific syllable [26].

The implication is direct. If behavioural phenotyping is to serve as a sensitive tool for detecting the effects of genetic background, age, and experimental perturbation, it must operate at multiple levels of the behavioural hierarchy simultaneously: the form of individual movement primitives, the statistics of their usage, and the higher-order structure of their sequential organisation. Each level may carry different, complementary information about organism state. Moreover, the analysis must be as free as possible from prior assumptions about which behaviours are relevant, precisely because the most interesting effects of a disease model or drug may be ones that have not yet been conceptualised. This freedom from bias, often disregarded in the final step of analysis, is crucial when

interpreting the retrieved behavioural space. The double requirement: sensitivity across hierarchical scales and minimal prior assumptions about behavioural content, is exactly the gap that the methods developed in this thesis aim to address, and it is the primary scientific motivation for applying them to the multi-strain, multi-age, multi-paradigm open-field dataset analysed in the following chapters.

Chapter 3

Data & Preprocessing

3.1 Arena Dataset

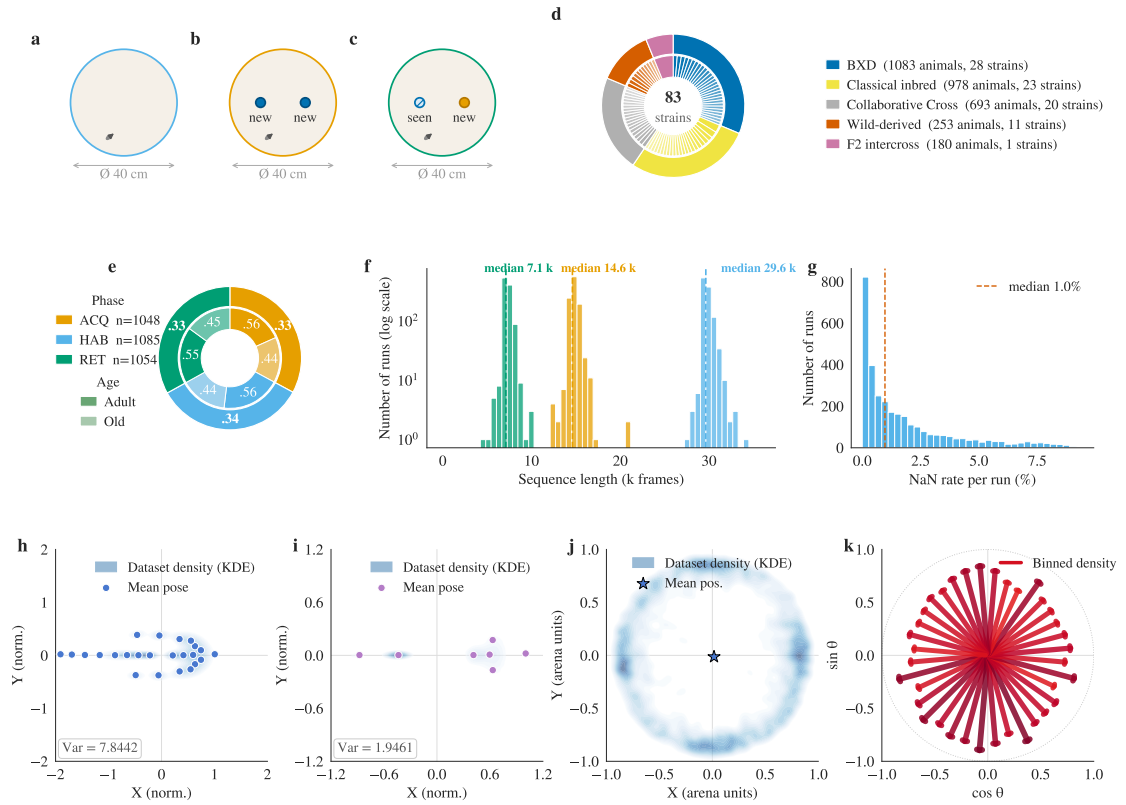


Figure 3.1: (a–c) Top view of the three task phases in a circular open-field arena (40 cm diameter): (a) habituation (no objects); (b) acquisition (two novel objects A, B, equidistant from the centre); (c) retention (one familiar object, one novel object C). (d) Strain composition ($n = 83$): genetic family (outer) and strain breakdown (inner, shaded by abundance). (e) Phase distribution across runs (outer) with age-group breakdown (inner; saturated: adult, light: old). (f) Sequence-length distribution per phase (log scale); dashed lines: per-phase medians. (g) Keypoint NaN rate per run; dashed line: median. (h–i) Mean pose over keypoint density (centre-aligned), without (h) and with (i) subsampling. (k–l) Distribution of animal positions (k) and heading directions (l) across all frames (centre-aligned); star: mean.

3.1.1 Experimental Design

The Arena dataset comprises recordings of the natural behaviour of 740 individual mice from 83 distinct strains (Fig. 3.1). Each individual underwent six recording sessions spanning three experimental phases, habituation, acquisition, and retention, each conducted at two distinct life stages: adult (4-8 months) and old (16-20 months) [10]. In total, 3,224 runs were collected, of which 3,187 were retained after quality filtering.

The habituation phase is a standard paradigm for studying natural, spontaneous locomotor behaviour, in which the mouse freely explores an empty arena [37]. In the acquisition phase, two novel objects are introduced equidistant from the arena centre, and in the retention phase one of those objects is replaced by a further novel one. These additional paradigms allow the assessment of memory-related behavioural changes as well as individual-object interaction patterns [7]. The three phases differ markedly in duration: habituation runs are the longest, reaching up to 30,000 frames (Fig. 3.1f). Recording each phase at two life stages allows the capture of ageing-related behavioural trajectories within a given strain [4]. Finally, among the 83 strains, some are more closely related genetically than others; they were therefore grouped into families, according to [5], to facilitate visualisation and analysis (BXD, Classical inbred, Collaborative Cross, Wild-derived, and F2 intercross; Fig. 3.1d) [10].

3.1.2 Preprocessing

The recorded videos were first processed with DeepLabCut [24], yielding time series of 27 2D keypoints per frame. The resulting keypoint tracks still contained missing values (NaNs), most commonly due to occlusions (Fig. 3.1g). Given the small overall proportion of missing data, a simple linear interpolation was applied. At training time, runs were divided into fixed-length windows; windows containing more than 1% NaN values were discarded as insufficiently reliable, while the remainder were retained and interpolated. To mitigate the pronounced body-size differences associated with both age and strain [10], each keypoint was standardised accordingly.

From the full 27-keypoint pose, a subset of 8 keypoints was selected: nose, neck, body centre, left ear, right ear, mid-dorsum, and tail base. Figure 3.1h-i shows the centre-aligned keypoints; the body centre is therefore omitted, yielding 7 keypoints. This selection was motivated by several factors. First, tail keypoints are frequently excluded from behavioural analyses, as they tend to reflect only a delayed echo of body movement and contribute little independent information [39]. Second, although keypoints were corrected for body size, residual strain- and age-related morphological differences remain detectable, particularly in the limbs, making limb keypoints a potential source of label leakage. Finally, this reduced keypoint set has demonstrated strong performance in the Keypoint-MoSeq pipeline [39, 26]. The keypoints were then centre-aligned by rotating each frame so that the neck-tail-base axis coincides with the x -axis, and subtracting the body centre position (Fig. 3.1k-l). The resulting distribution shows a broadly uniform heading direction and a markedly peripheral positional distribution within the arena, consistent with the wall-hugging behaviour commonly observed in rodents [42].

In a final step, the model was trained solely on the centre-aligned keypoint coordinates, discarding absolute positional and rotational features. This choice, motivated by the embedding quality it yields, is discussed further at the beginning of the Results section. Taken together, all preprocessing steps were designed to remove non-behavioural confounds from the keypoints that could bias the learned representations toward a specific label (strain, age, or experimental phase).

Out of the 83 strains, 80 strains were kept. B6CAST-129SPWK-F2, with 180 recordings, was discarded because it is an F2 intercross rather than an isogenic strain: as a genetically segregating population it shows far greater within-group variability than the inbred strains and cannot be treated as representative of a single strain [8]. Additionally, JF1/MsJ and PANCEVO/EiJ were also excluded from the analysis due to too few recordings.

3.2 Benchmark Datasets

The following section describes the two datasets used for comparison and validation of the proposed implementations: Shot7M2 and MABe22.

3.2.1 Shot7M2

Shot7M2 is a synthetic, hierarchically annotated keypoint dataset comprising 4,000 episodes of 1,800 frames each, generated at 30 Hz [34]. Each frame consists of 26 keypoints in 3D, describing the pose of a single basketball player. The dataset encompasses three nested levels of behavioural hierarchy across both spatial and temporal scales: movemes, actions, and activities. Labels are often body-part-specific, for example, left-foot and right-foot movemes encode foot-ground contact events, and multiple labels can co-occur within the same frame. Crucially, the hierarchy is strictly nested: each level subsumes the one below it, with activities spanning entire episodes and the entire set of keypoints, actions ranging from 12 to 400 frames, and movemes from 3 to 20 frames. This strong compositional spatio-temporal structure, combined with the precise ground-truth annotations afforded by synthetic generation, makes Shot7M2 a well-controlled benchmark for evaluating models designed to learn hierarchical behavioural representations [34]. In this study, it is used to validate and compare the proposed model extensions.

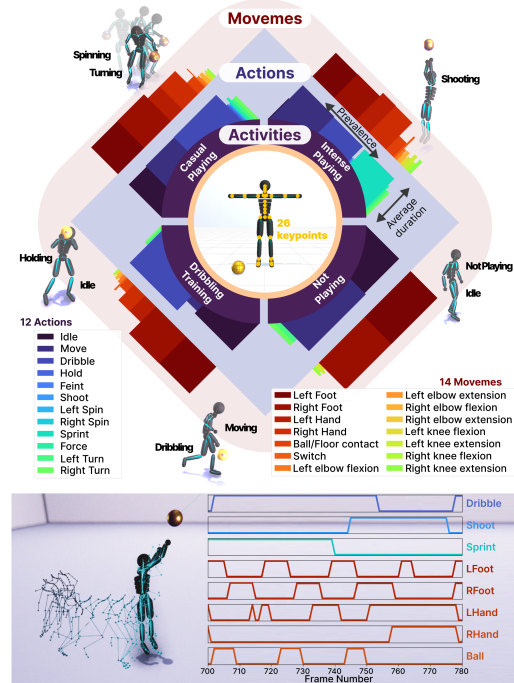


Figure 3.2: The synthetic basketball dataset (7.2M frames, 26 3D keypoints) spans three levels of behavioural hierarchy: movemes (fine-grained, sub-second), actions (intermediate), and activities (coarse, episode-level), comprising 14 movemes, 12 actions, and 4 activities. Reproduced from Stoffl et al. [34], licensed under CC BY 4.0.

3.2.2 MABe22

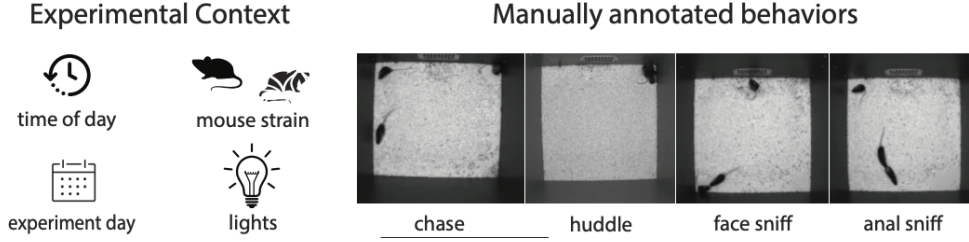


Figure 3.3: Three mice per clip are tracked via 12 2D keypoints. Sequence-level and frame-level behavioural labels capture both experimental metadata and social interactions. Reproduced from Sun et al. [36]

MABe22 is a multi-species behavioural dataset introduced as part of the MABe 2022 benchmark challenge [36], including mouse, ant, and fly keypoint recordings. In this study, only the mouse subset was used, comprising 5,336 clips of 1,800 frames each, all recorded at 30 Hz. A distinctive feature of this dataset is its multi-animal design: each clip captures three mice simultaneously, enabling the study of social interactions [3]. Each animal is tracked via 12 anatomically defined 2D keypoints. Behavioural annotations are provided at two granularities: sequence-level labels encoding experimental metadata (experiment day, strain, and lighting condition) and frame-level labels capturing social behaviours (approaches, chases, close proximity, physical contact, huddling, oral-ear contact, oral-genital contact, oral-oral contact, and watching). Unlike Shot7M2, MABe22 does not impose an explicit hierarchical organisation, making it a complementary benchmark for evaluating representations of naturalistic, unconstrained social behaviour, in addition to being a more relevant comparison to rodent natural behaviour present in the arena dataset.

Chapter 4

Methods

4.1 hBehaveMAE

4.1.1 Architecture

hBehaveMAE [34], described in Section 2.4.3, was adopted as the core architecture for this work. All experiments build on the multi-scale decoding variant with fusion head. Prior to any extension, the reported results of Stoffl et al. were reproduced on Shot7M2 and MABe22, with deviations of at most -1.44 percentage points, consistent with random seed variation and establishing a reliable baseline.

Reconstruction vs. representation trade-off. The hBehaveMAE architecture presents two related challenges. The first is a duality of objectives: the model must learn to encode pose trajectories in a form that supports robust reconstruction under high masking ratios, while also producing embeddings that capture behaviourally meaningful structure. The balance between these objectives is governed by the reconstruction difficulty. If the task is too easy, due to a low masking ratio or an overly powerful decoder, the encoder need not learn any compact or informative representation. Conversely, if the task is too difficult, the model fails to learn a meaningful compressed representation altogether.

The second challenge concerns the hierarchical structure of the embeddings. While Hiera’s pooling architecture may produce representations that differ across stages as a side effect, only the final stage is directly tied to the reconstruction objective under single-scale decoding. The multi-scale decoding strategy was introduced precisely to address this, by connecting all stages to the reconstruction loss. It has, however, shown limited performance gains to date. This study therefore proposes to strengthen this strategy through hierarchical nested dropout, a method that has already demonstrated promising results in hierarchical visual transformers [2]. The key motivation is that the current mask unit masking scheme applies a uniform masking difficulty across all encoder levels, which ineffectively exploits the reconstruction capacity of lower-hierarchy stages.

Finally, hBehaveMAE was originally designed for fixed-length time series. However, as the Arena dataset illustrates, naturalistic behaviour is often variable in duration and may unfold over long timescales. Fixing the temporal input dimension requires the model to learn absolute temporal positional embeddings, constraining inference to sequences of a specific length. To overcome this limitation, a length-agnostic implementation of hBehaveMAE was developed, enabling the model to encode sequences of arbitrary length.

This study introduces three extensions to the original architecture: hierarchical nested dropout to enforce inter-level representational specialisation (Section 4.2), a length-agnostic

implementation enabling variable-length inference (Section 4.3), and a full-sequence embedding extraction strategy (Section 4.4.2).

4.1.2 Model Configurations

Table 4.1: Model configurations across datasets.

Dataset	Input size	Stages	Q strides	Patch kernel	Out embed dims	Decoding strategy	Mask ratio
Arena	$900 \times 1 \times 14$	2, 3, 4	2,1,1; 5,1,1	$1 \times 1 \times 14$	78, 128, 256	multi	0.80
MABe22	$900 \times 3 \times 24$	3, 4, 5	5,1,1; 1,3,1	$3 \times 1 \times 24$	128, 192, 256	multi	0.875
Shot7M2	$400 \times 1 \times 72$	2, 3, 4	2,1,4; 2,1,6	$2 \times 1 \times 3$	78, 128, 256	single	0.70

The MABe22 and Shot7M2 configurations were taken directly from the best-performing models reported in Stoffl et al. [34]. For the Arena dataset, no extensive hyperparameter search was conducted; instead, an initial configuration was selected based on principled choices described below.

The temporal window was set to 900 frames to match the MABe22 setup, and the window size appeared to have limited impact on downstream performance (discussed further in the Results section). The input feature dimension encodes the 7 centre-aligned 2D keypoint coordinates without positional information, yielding a $900 \times 1 \times 14$ input. Given the large scale of the Arena dataset and the goal of an initial tractable configuration, a lightweight model was adopted, following the stage structure and embedding dimensions of Shot7M2. The query strides were set to pool at most 10 frames (≈ 400 ms at 25 Hz), corresponding to the median behavioural syllable duration reported in Keypoint-MoSeq analyses [39].

Masking ratio and mean loss of contextual information. The masking ratio is the most impactful hyperparameter in this setting. Unlike video MAEs, which typically mask individual tokens at ratios above 0.9, hBehaveMAE masks at the mask unit level. Depending on the temporal extent of the mask unit, determined jointly by the patch kernel and pooling strides, a single masked unit can occlude a long range of consecutive frames, potentially eliminating all local temporal context. To quantify this, we define the *Mean Loss of Contextual Information* (MLCI) as the expected number of consecutive frames masked within a visible context window:

$$\text{MLCI} = \frac{\text{mask unit length} \times \text{masking ratio}}{1 - \text{masking ratio}}. \quad (4.1)$$

For MABe22, the temporal mask unit spans 15 frames after all pooling and patching operations, yielding an MLCI of 105 frames (3.5 s at 30 Hz). For Shot7M2, the corresponding MLCI is 18.6 frames (≈ 600 ms at 30 Hz). These values set the range of contextual disruption within which the model has shown strong results. For the Arena dataset, the masking ratio was set to 0.80, yielding an MLCI of 40 frames (1.6 s at 25 Hz), between the Shot7M2 and MABe22 bounds. This configuration is not claimed to be optimal; a systematic search over masking ratios, pooling strides, and model depth would be a natural extension of this work, but was outside the scope of this study.

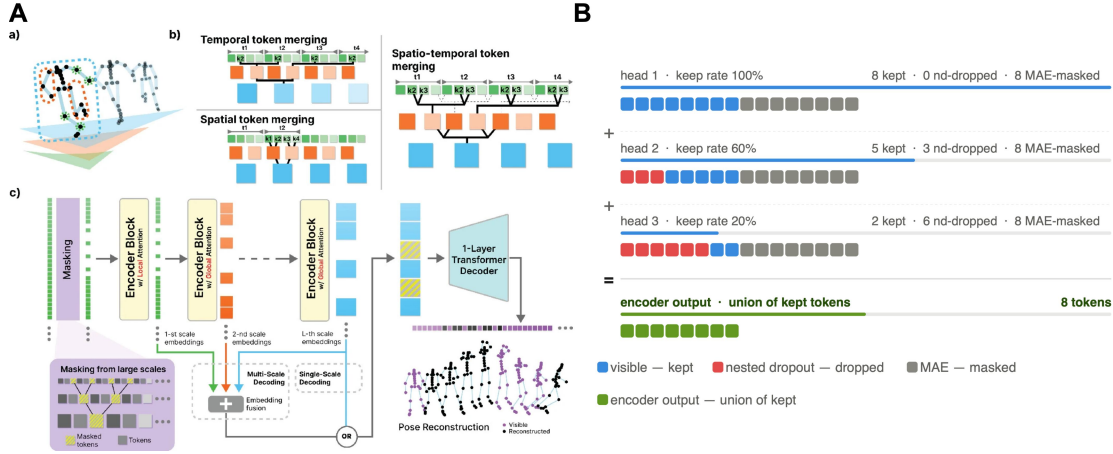


Figure 4.1: **(A)** The hBehaveMAE framework. **(a)** Pose trajectories (2D/3D, one or several individuals) are patched over spatial and/or temporal components — e.g. a full pose over multiple frames (blue) or a single keypoint at one timestep (green) — then linearly embedded to form the input token sequence. **(b)** During the forward pass, tokens are merged temporally, spatially, or spatio-temporally to build a predefined hierarchy. **(c)** A random subset of tokens is masked at the highest scale and back-projected to the lowest level; successive encoder blocks (local or global attention) fuse tokens at each scale, and a single-layer transformer decoder reconstructs the masked pose coordinates from either multi-scale or single-scale embeddings. **(B)** Proposed nested-dropout token-keeping across attention heads. Each head retains a nested subset of the input tokens at a decreasing keep rate (head 1: 100%, 8/8 kept; head 2: 60%, 5 kept, 3 dropped by nested dropout; head 3: 20%, 2 kept, 6 dropped), while the remaining tokens are MAE-masked. The encoder output is the union of the tokens kept across heads. Colour code: blue, visible (kept); red, dropped by nested dropout; grey, MAE-masked; green, encoder output (union of kept tokens). Panel **(A)** reproduced from Stoffl et al. [34], licensed under CC BY 4.0.

4.2 Nested Dropout

Nested dropout has already demonstrated promising results in image encoding with FlexTok [2]. Its application was, however, constrained to the single-layer, flat embedding space produced by that pipeline. In FlexTok, hierarchy emerges horizontally along the embedding dimensions, whereas hBehaveMAE constructs hierarchy vertically, across distinct encoder levels. The nested dropout scheme must therefore follow the same vertical direction.

Due to the restrictive mask unit organisation and pooling stages of hBehaveMAE, it is not possible to introduce dropout either at the token level or within the encoder itself. Doing so would either break the attention mechanism or cause missing tokens to be pooled and propagated upward through the hierarchy, ultimately producing the same uniform masking difficulty across all layers that the method aims to avoid.

To affect each encoder level independently, nested dropout is instead introduced within the fusion head. At this stage, a distinct representation is available for each level and can be modified in isolation. Concretely, the implementation defines a minimum fraction of mask units to retain per level, and then interpolates the per-level keep rates, either linearly or exponentially, between this minimum and 1.0 at the final level, ensuring that at least one encoded mask unit is always present for the summation step. Two strategies were considered for increasing the reconstruction difficulty: replacing dropped units with a learned mask token, or zeroing them out entirely. Although the former is more intuitive, the latter yielded better results in practice and was therefore adopted. To compensate for the reduction in signal magnitude caused by unit dropping, each level’s contribution is rescaled by $1/k$, where k is the corresponding keep rate [33, 31].

Dropout direction. A natural question is whether the nesting should proceed from fine to coarse levels, or in the reverse direction. In FlexTok, this choice doesn’t occur, since no prior hierarchy exists in the flat embedding space and hierarchy emerges entirely from the nested dropout itself. The situation in hBehaveMAE is different, as two forms of pre-existing hierarchy are present. The first arises even in non-hierarchical models, as a consequence of the well-known progressive abstraction effect of sequential attention layers [30]. The second is structurally imposed by the pooling stages, which define a monotonically coarsening spatio-temporal resolution gradient from the first to the last encoder level.

Following the FlexTok convention of nesting from fine to coarse would mean always retaining at least one mask unit from the coarsest, highest-level layer. This would effectively create a continuum between the single-scale decoding strategy, where only the last layer contributes, corresponding to a minimum keep rate of 0.0, and the multi-scale strategy, where all layers contribute equally, corresponding to a minimum keep rate of 1.0. This direction is consistent with the second form of pre-existing hierarchy, in which the model primarily relies on coarse-resolution embeddings for reconstruction and progressively refines them using fine-resolution contributions. However, fine levels are also earlier levels, and they have access to fewer sequential attention blocks and less representational capacity, making this additional burden potentially counterproductive.

This choice ultimately hinges on a key assumption about the multi-scale decoding strategy: does the decoder rely more heavily on coarse or fine encoder levels? One might expect fine levels to be more directly useful, as their representations are closer in resolution to the raw keypoint input. However, in the fusion head, fine-level representations are upsampled to match the coarser resolution of the last stage before being summed, thereby losing their resolution advantage while still paying the cost of limited sequential attention. This reasoning suggests that the decoder primarily relies on the last, coarsest encoder level for reconstruction. Therefore, to maximally increase the reconstruction difficulty at the levels that contribute least, the nesting direction should proceed from coarse to fine, retaining the finest levels with the highest probability and progressively dropping the coarser ones.

4.3 Length-agnostic implementation

The past implementation of hBehaveMAE fixed the temporal scale of the input effectively dictating the maximum possible attention window of the model (For example 400 frames for Shot7M2 models and 900 frames for MABe22 models). Doing so, allowed the model to learn temporal positional embeddings for each patched frame, however it also leads to two major downsides. First, as discussed it is an additional hyperparameter of the model dictating a possible important feature of attention. Secondly, at inference time, it forces one to divide the behavioural time series into chunks in order to obtain a full representation.

The only parameter preventing variable-length inference in hBehaveMAE is the learned temporal positional embedding, as all other model parameters are inherently length-agnostic. Most modern language transformers achieve length-agnosticism by replacing learned absolute positional embeddings with Rotary Position Embeddings (RoPE) [35], which encode position implicitly within the attention mechanism by rotating query and key vectors by an angle proportional to their position in the sequence. This eliminates the need to fix the sequence length at training time and allows the model to generalise across context lengths at inference.

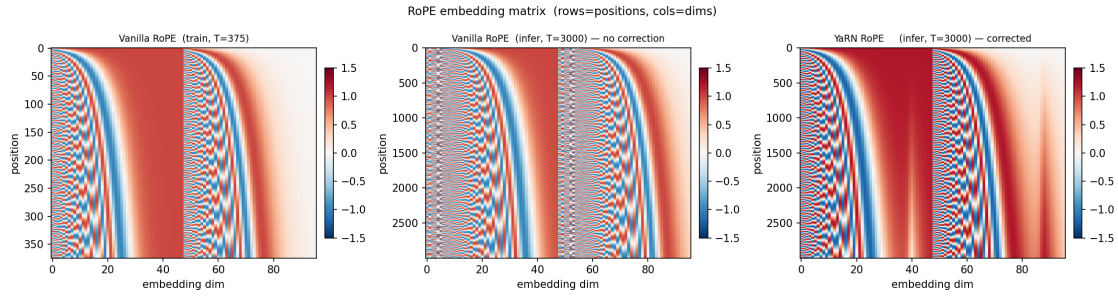


Figure 4.2: Each panel shows the rotary positional embedding as a heatmap with token positions along the rows and embedding dimensions along the columns; colour encodes the embedding value. Low dimensions (left) rotate at high frequency and produce fine oscillations, while high dimensions (right) rotate slowly and vary smoothly. **(Left)** Vanilla RoPE at the training context length ($T = 375$), showing the well-behaved banded structure. **(Middle)** Vanilla RoPE extrapolated to an inference length of $T = 3000$ without correction: the high-frequency dimensions alias beyond the trained range, producing the noisy bands and degrading the regular structure. **(Right)** YaRN-corrected RoPE at the same inference length ($T = 3000$): rescaling the rotation frequencies restores a smooth, regular pattern across the full extrapolated range.

A known limitation of standard RoPE, however, is that when training is performed on a fixed window length, the model is exposed only to a bounded range of rotational angle values. Feeding longer sequences at inference time produces out-of-distribution positional encodings that the model has never encountered, often leading to degraded performance or systematic biases (Fig. 4.2). YaRN [27] addresses this by rescaling the rotation frequencies proportionally to the ratio between the inference context length and the training context length, effectively remapping out-of-distribution positional encodings back into the range seen during training. This implementation was adopted here to enable robust inference over sequences longer than the training window.

Sinusoidal positional embeddings defined within mask units were also investigated as an alternative. Encoding position only within the span of a single mask unit has the advantage of being fully context-invariant, as the embedding values are independent of the absolute position in the sequence. However, this approach has a critical failure mode: at the coarsest encoder level, each mask unit collapses to a single token, at which point the intra-unit positional embedding carries no information whatsoever. Because this would result in a complete absence of temporal structure at the highest hierarchical level, this approach was discarded, despite showing some promise in the context of upscale fusion heads explained later in the thesis.

4.4 Embedding Extraction Strategies

4.4.1 Windowed Extraction

The main interest of hBehaveMAE resides in the embeddings obtained at the bottle neck of the reconstruction task between the encoder and decoder. Once a model is trained, they are obtained by feeding a window through the encoder module and collecting the representation at the end of each stage. A stage being defined as a series of block operating at a specific spatio-temporal dimension, with pooling applied between each stage. A common approach in the field and used in unsupervised learning methods like Keypoint-Moseq, Ts2Vec, BAMS is to obtain a per-frame embeddings. To do so one window is built centred on every frame of sequence of interest. The middle token of the obtained

embeddings is then collected for each layer for a given centre frame. Additionally Stoffl et al. introduced bootstrapping by padding the window of length l at both edges with $l/2$ and collecting embeddings for all windows with a stride 1. The collected embeddings are then averaged to obtain the final representation. While the last layer has all dimensions pooled resulting in a single embedding per temporal resolution, this is not the case of earlier layers which can have additional spatial dimensions. These extra dimensions are either flattened or summed. According to Stoffl et al. method MABe22 embeddings are summed while Shot7M2 are flattened.

4.4.2 Full-Sequence Extraction

The windowed extraction method leads to dense information embedding but also very correlated adjacent frames, in addition to require a consequent amount of padding at edges and a significant compute time. In this regard an additional extraction strategy was implemented leveraging variable length prediction capabilities of the length agnostic models. This method, much simpler, simply runs a given time series of any length through the encoder module of a trained model and collect embeddings at the end of each stage. It then flattens extra dimensions. The resulting embeddings are not at the frame level but at the temporal resolution defined by the pooling stages. This essentially divides our time series in discrete mask unit sequences. In order to reduce the effect of this discrete sectioning a similar bootstrapping as previous extraction method is applied but with a padding of mask unit size ensuring the representation is not bound to the mask unit centering. All collected representation with a stride 1 are then averaged. Finally, to match the previous extraction method embeddings are repeated to match the per frame embedding resolution. This addition will allow to compare the two extraction methods.

4.4.3 Pooled And Fused Embeddings

Two additional sub-methods were implemented, on top of the windowed extraction, to further evaluate the embeddings of Stoffl et al. extraction strategy. The first one, pooling-extraction simply generates all subsequent layers by average pooling the first layer according to the defined pooling stage of the model. In essence, it ablates all transformation learned in the model past the first layer and simply matches the spatio-temporal resolution of normally extracted embeddings. This method will allow to test the importance of the spatio-temporal resolution compared to the actually learned transformation. As discussed in the hBehaveMAE architecture section, the multi-decoding strategy learns a pooling of each stage to match the last stage before summing them and feeding them to the decoder. In order to study the effect of this learned pooling on the resulting embeddings, the second sub-method was implemented. This method simply collects the fused embeddings of each layer just before the summation and then proceed to the same bootstrapping pipeline as explained before.

4.5 Evaluation Framework

This section describes the approaches used to evaluate the behavioural relevance of the extracted representations. In the context of hBehaveMAE, this step is crucial: as discussed in the architecture section, an insufficiently difficult reconstruction task could result in representations that are little more than a trivial transformation of the raw input, encoding no behaviourally meaningful structure. Three complementary evaluation strategies

are employed. Linear probing assesses how well the representations predict labels of interest. Behavioural segmentation and clustering evaluates whether the embedding space organises into discrete, interpretable behavioural units. Finally, Centered Kernel Alignment (CKA) and layer ablation analysis are used to probe the internal structure of the encoder, characterising the similarity between layer representations and the unique contribution of each stage to the reconstruction objective.

4.5.1 Linear Probing

The most common approach in the field is linear probing. It consists in evaluating the predictability capabilities of the collected representation given labels of interest. In this study, two implementations will be discussed : per-frame linear probing and per-sequence linear probing.

Per-Frame Linear Probing The first method was implemented by Stoffl et al. in the context of MABe22, Shot7M2 and hBabel datasets which all include per-frame labels. Following the MABe22 challenge protocol, a linear classifier is trained independently for each label using frozen representations. The linear classifier is implemented as a Ridge regression for continuous labels (evaluated with MSE) or a balanced-weight RidgeClassifier for discrete labels (evaluated with macro F1 score). To reduce variance, training and evaluation are repeated across multiple random seeds, and predictions from models trained at different regularisation strengths ($\alpha \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$) are ensembled by averaging the per-seed predictions before computing the final score. The reported score is thus a pooled frame-level F1 computed over all frames of both the public and private test splits after seed-level ensembling. This method is also applied to sequence-level label tasks, where the same frame-level classifier is trained to predict a sequence-level label from each individual frame’s representation, treating the sequence label as if it were replicated at every timestep. The final prediction is obtained by averaging the per-frame predictions of the linear probe within each clip and applying a threshold $\tau = 0.5$ to produce a binary sequence-level decision. The F1 score is then computed at the clip level. This script was adapted for multi-task prediction in the context of the arena dataset for metadata variables such as strain or experimental phase which admit more than two categories. In this context a vectorised majority vote is applied across seeds at each frame: per-seed predictions are one-hot encoded, summed across seeds, and the class with the highest cumulative vote is selected as the final prediction. The resulting multi-class predictions are then evaluated with macro-averaged F1. This method is evaluated on a fixed test set for Shot7M2 and MABe22, and one was thus generated for the Arena dataset including a randomly sampled subject of each available strain and all its 6 videos including the 2 age phases and 3 experimental stages. Additionally, the syllable labels obtained from previous Keypoint-MoSeq analysis on the dataset were used as alternative per-frame labels since all other labels are per-sequence.

Per-Sequence Linear Probing The second method was implemented by Arash Sal Moslehian to study Keypoint-Moseq syllables expression per sequence. The input feature to the probe is not a per-frame embedding but a sequence-level representation for each recording. The probe is a multi-class logistic regression with ℓ_2 regularisation, fitted inside a `StandardScaler` $\rightarrow$ `LogisticRegression` pipeline. Generalisation is estimated with repeated stratified group k-fold cross-validation (k=5 folds, 5 repeats), where grouping is done by subject identifier to ensure that no individual animal appears in both the training

and test sets of the same fold. This is critical in a multi-strain setting where the goal is to measure strain-level rather than individual-level decodability. Across all folds and repeats, out-of-fold (OOF) predicted probabilities are accumulated and averaged, and the final hard prediction for each recording is derived from the averaged probabilities. Performance is reported as pooled OOF accuracy. To obtain per-sequence representation, the embeddings from hBehaveMAE were simply averaged over all frames.

4.5.2 Behavioural Segmentation and Clustering

One of the advantages of hBehaveMAE lies in the continuous nature of its behavioural embeddings. However, it is common practice in unsupervised behavioural analysis to segment this continuous behavioural space into discrete units [23, 39]. Methods such as VAME [22] and B-SOiD operate on this principle, and discretising the embedding space also enables a direct comparison with Keypoint-MoSeq [39], which is inherently discrete by design. To discretise the continuous behavioural space k -means algorithm was used. Clustering was applied either to per-frame embeddings directly (see supplementary Figure 1), or to per-sequence aggregated embeddings. In both settings, and in order to match the number of behaviourally relevant syllables recovered by the Keypoint-MoSeq analysis, the number of clusters was fixed to 50.

4.5.3 Centered Kernel Alignment & Layer Ablation

hBehaveMAE, like many deep learning models, can be difficult to interpret. It is, however, crucial to understand its internal learning mechanisms in order to obtain meaningful embeddings. To this end, two complementary analyses were conducted: representational similarity between latent representations was studied via Centered Kernel Alignment (CKA), and the unique contribution of each encoder scale to the reconstruction objective was assessed through a layer ablation study.

Centered Kernel Alignment. CKA [19] has become a widely used tool for probing the internal representations of neural networks, enabling comparisons both within and across architectures in a scale-invariant manner. Given two representation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$, linear CKA is defined as:

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}, \quad (4.2)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The measure is bounded in $[0, 1]$ and equals 1 when the two representations are linearly isomorphic, providing an intuitive notion of similarity that is invariant to orthogonal transformations and isotropic scaling. In this work, linear CKA is computed between all pairs of encoder levels across the hierarchical blocks, as well as between a layer and the final fused representation obtained after the fusion head.

Layer Ablation. Representational similarity via CKA provides insights into how internal representations relate to one another, but does not directly characterise each scale’s contribution to the model’s reconstruction objective. To address this, a layer ablation study was implemented. The multi-scale decoding in hBehaveMAE fuses encoder outputs from all hierarchical levels via an additive summation before the decoder. The contribution of each scale can therefore be isolated cleanly by zeroing out the corresponding head output before fusion, which is equivalent to removing that term from the sum

while leaving all remaining heads intact and in-distribution. Two ablation strategies were evaluated. In the *zero-ablation* condition, a given scale’s contribution is set to zero. In the *mean-replacement* condition, the zeroed contribution is replaced by the dataset-mean embedding of that head, computed over 1,000 batches on the Shot7M2 dataset. The latter provides a less disruptive baseline, as it replaces the ablated signal with a constant neutral representation rather than a zero vector, which could introduce an artificial bias and was thus favoured in all consequent analysis. The latter strategy was evaluated over 1,000 batches using fixed, pre-generated masks shared across all conditions so that stochastic masking cannot confound comparisons between conditions. Three families of conditions were tested: (i) a *baseline* condition in which all scales are active; (ii) *single-scale isolation*, in which only one scale k is active while all others are ablated; and (iii) *leave-one-out*, in which all scales are active except scale k . Results are reported as mean reconstruction MSE across a range of training epochs.

4.6 Deep Baselines

4.6.1 TS2Vec

TS2Vec is a framework for learning hierarchical representations of time series through contrastive learning [43]. Unlike auto-encoder-based approaches such as VAME or BehaveMAE, TS2Vec applies contrastive losses at multiple temporal scales, encouraging consistency across resolutions and allowing hierarchical structure to emerge without explicit architectural pooling constraints apart from simply temporal convolutional kernels.

Several embedding extraction strategies are proposed by Yue et al.; in this study, the bootstrapped sliding-window extraction was used, with a window size of 900 frames matching the training configuration. Full-sequence extraction — which pools representations down to a single embedding per sequence — was also evaluated but found to be unsuitable for this dataset: since the temporal pooling levels required for full sequences were never observed during training on fixed-length windows, the resulting embeddings were heavily biased by input length and differed substantially across experimental phases, making them unreliable for behavioural analysis.

The model was trained with 900 frames input length, 60’000 samples and a batch size of 512. The model’s output dimensions was set to 128 with 256 hidden dims and a depth of 10 and it was trained for 20 epochs. The number of training samples was capped at 60’000 due to a memory constraint in the current TS2Vec training script, which requires the full dataset to be instantiated in memory at once. Given the size of the Arena dataset, this limit ensures approximately 10 sampled windows per recording, providing reasonable coverage while remaining within memory bounds.

4.6.2 TCN

As an additional deep baseline, a Temporal Convolutional Network (TCN) autoencoder was implemented, inspired by the long-term branch of the BAMS architecture [1]. The encoder consists of a stack of dilated residual blocks, each comprising two Conv1d layers with the same dilation factor, layer normalisation, GELU activations, and a residual skip connection. Dilation increases exponentially across blocks, yielding a receptive field that grows as $d_{\text{base}}^{n_{\text{layers}}} \times k$, where d_{base} is the dilation base, n_{layers} the number of blocks, and k the kernel size. Following the encoder, a global average pooling operation over the temporal dimension produces a fixed-size sequence embedding regardless of input length, making the model length-agnostic by construction. The decoder mirrors the encoder

with reversed dilation order, broadcasting the pooled embedding back over the temporal dimension before reconstructing the full per-frame pose sequence; the training objective is MSE reconstruction over all frames.

The model was trained on the centre-aligned, no-position Arena keypoints with the following configuration: window size of 900 frames, TCN channels [64, 64, 64, 32], embedding dimension 64, kernel size 2, dilation base 2 (receptive field ≈ 32 frames, ≈ 1.3 s at 25 Hz), dropout 0.1, batch size 512, learning rate 10^{-4} with cosine annealing, and weight decay 0.05, trained for 200 epochs. Sequence-level embeddings were obtained by global average pooling over the temporal dimension of the encoder output, and evaluated using the per-sequence linear probing protocol.

Chapter 5

Results

5.1 Benchmark Datasets

5.1.1 Synthetic Hierarchical Structure Drives Hierarchical Representation Learning in hBehaveMAE

The results of Stoffl et al.’s hierarchical and non-hierarchical BehaveMAE models were reproduced on the Shot7M2 and MABe22 datasets as a baseline for further implementations [34]. The reproduced F1 scores deviated from the published values by at most -0.57 percentage points for the non-hierarchical Shot7M2 model, -0.08 for the hierarchical Shot7M2 model, -1.44 for the non-hierarchical MABe22 model, and -1.22 for the hierarchical MABe22 model. These minor deviations are consistent with differences in random seed initialisation and are considered sufficiently close to establish a reliable baseline (See Supplementary Table 1, 2, 3 for detailed results). Although Stoffl et al. did not report exact numbers for the multi-scale decoding strategy, they described it as performing “on par” with the single-scale baseline; this result is reproduced here. On Shot7M2, the single-decoding model achieves best-layer F1 scores of 0.668, 0.581, and 0.809 for movemes, actions, and activities respectively, compared to 0.664, 0.572, and 0.799 for the multi-scale variant. On MABe22, the single-decoding model obtains per-frame and per-sequence F1 scores of 0.218 and 0.806, against 0.216 and 0.789 for multi-scale decoding (Fig. 5.1B,E) [34]. Stoffl et al. did not report per-level F1 curves for MABe22, noting the absence of meaningful hierarchical structure in that dataset; our results support this finding, with mean per-layer F1 scores remaining largely stable across levels (0.457, 0.512, 0.441).

To understand why hBehaveMAE trained on Shot7M2 develops a clear hierarchical representational structure while the model trained on MABe22 does not, the temporal properties of the two datasets were examined. Power Spectral Density analysis reveals a fundamental distinction: Shot7M2 contains significant periodic components at short timescales (3–100 frames), whereas MABe22 follows a characteristic $1/f$ spectral profile with no prominent temporal structure, consistent with naturalistic and unconstrained behaviour (Fig. 5.1A,D). Quantitatively, Shot7M2 exhibits a markedly steeper spectral decay ($\beta = 2.95$) than MABe22 ($\beta = 0.78$). This contrast reflects the top-down generative process underlying Shot7M2, which concentrates energy at specific timescales by construction, whereas MABe22 displays the relatively flat, scale-free temporal structure characteristic of naturalistic animal behaviour.

To further investigate this difference, the variance explained (η^2) by each behavioural label was computed for both datasets and correlated with the per-label increase in F1

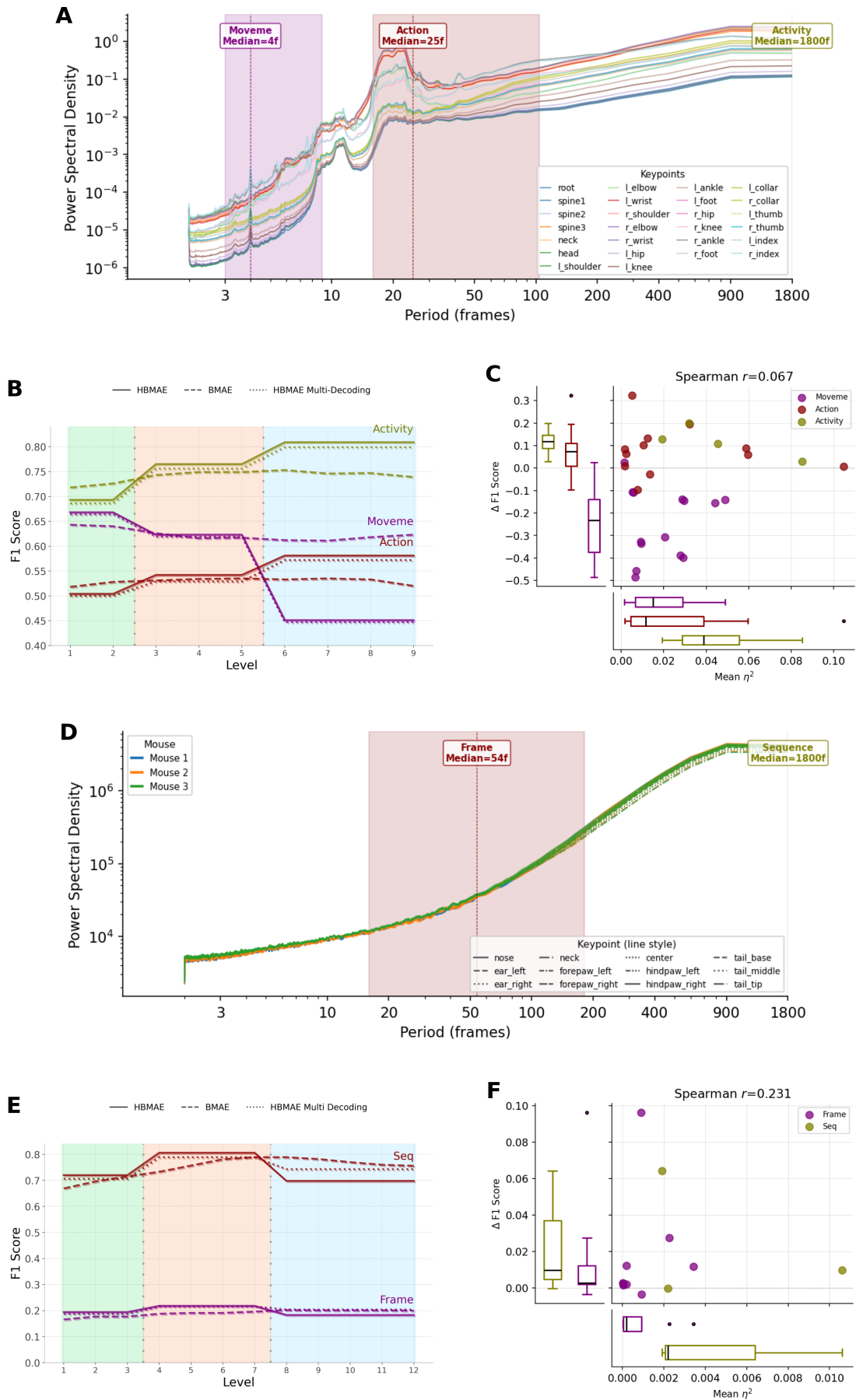


Figure 5.1: Top row: Shot7M2; bottom row: MABe22. **(A)** Power spectral density of the Shot7M2 key-point trajectories as a function of period (in frames). Shaded bands mark the three behavioural timescales (moveme, action, and activity; vertical lines indicate their median periods of 4, 25, and 1800 frames). The spectrum shows pronounced periodic components at short timescales (3–100 frames) and a steep decay ($\beta = 2.95$). **(B)** Best-layer F1 score per encoder level on Shot7M2 for the hierarchical model (HBMAE, solid), the non-hierarchical model (BMAE, dashed), and hierarchical multi-scale decoding (dotted), for movemes (purple), actions (red), and activities (olive). Background shading denotes the hierarchical blocks; lower levels best decode movemes while higher levels best decode activities. **(C)** Per-label increase in F1 score between the first and last encoder layers ($\Delta F1$) versus the variance explained by each label (mean η^2) on Shot7M2, coloured by behavioural level, with marginal box plots. Spearman $r = 0.067$. **(D)** Power spectral density of the MABe22 mouse-triplet trajectories (one colour per mouse). Shaded bands mark the frame- and sequence-level timescales (median 54 and 1800 frames). In contrast to Shot7M2, the spectrum follows a shallow, near- $1/f$ profile ($\beta = 0.78$) with no prominent temporal structure. **(E)** Best-layer F1 score per encoder level on MABe22 for the frame-level (purple) and sequence-level (red) tasks; line styles as in (B). Scores remain largely flat across levels, indicating little hierarchical structure. **(F)** $\Delta F1$ versus mean η^2 per label on MABe22, coloured by task, with marginal box plots. Spearman $r = 0.231$.

score between the first and last encoder layers. Labels in Shot7M2 show substantially higher explained variance (up to $\eta^2 = 0.083$) than those in MABe22 (up to $\eta^2 = 0.010$), and the two datasets also differ in the relationship between label variance and layer-wise F1 improvement: the Spearman correlation between η^2 and $\Delta F1$ is $r = 0.067$ for Shot7M2 and $r = 0.231$ for MABe22 (Fig. 5.1C,F). Notably, MABe22 labels with very low explained variance ($\eta^2 < 0.001$) show negligible layer-wise improvement ($\Delta F1 < 0.02$), suggesting that the absence of hierarchical structure in the MABe22 embeddings is directly related to the low behavioural variance captured by its labels, rather than a limitation of the model architecture itself.

The spatial resolutions of the hBehaveMAE encoder levels trained on Shot7M2, covering 3, 12, and 72 keypoints respectively, corresponding to a single 3D keypoint, a limb in 3D, and a full subject in 3D, were designed to match the hierarchical spatial structure of the dataset, where movemes involve few keypoints, actions involve more, and activities span the entire body. However, the pooled embedding results show that the prediction power of each level cannot be explained by its spatio-temporal resolution alone (Fig. 5.2A). Activity and action F1 scores remain relatively stable across pooling levels (0.504 and 0.491 for actions, 0.693 and 0.676 for activities), while moveme prediction decreases substantially, from 0.668 to 0.517, closely mirroring the pattern observed in the trained hierarchical layers (0.668 and 0.451). This suggests that the decrease in moveme prediction of higher layers is at least partially due to their spatio-temporal resolution while the increase in action and activity prediction cannot be solely explained from it.

To isolate the contribution of the learned pooling in the fusion head from that of the transformer blocks themselves, the pre-fused embeddings of each layer, collected just before the summation step, were extracted and evaluated at the last layer’s resolution, allowing a direct comparison of the behavioural information encoded by each level. As expected, the last layer’s pre-fused embeddings match the multi-scale last layer’s results (Fig. 5.2B). Interestingly, the activity and action predictions of the pre-fused second layer (red) match their unfused counterparts, while the first layer actually improves after pre-fusion, reaching F1 scores of 0.524 and 0.709 compared to 0.500 and 0.686 without fusion. This improvement is a direct signature of the reconstruction objective: the fusion head is explicitly linked to the decoder loss, whereas the earlier transformer blocks are only connected to the reconstruction through the final layer. Notably, moveme prediction is consistently hurt by the learned pooling across all levels, consistent with the finding that

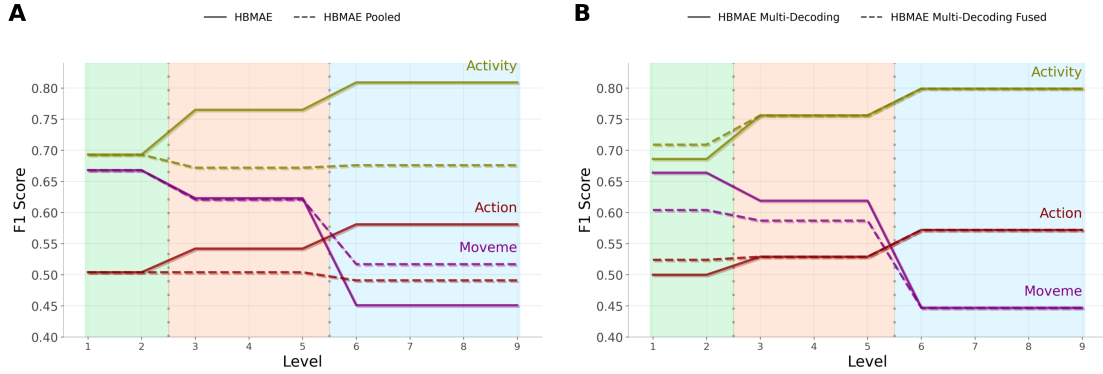


Figure 5.2: Both panels show the best-layer F1 score per encoder level for movemes (purple), actions (red), and activities (olive); background shading denotes the three hierarchical blocks, whose spatial resolutions cover 3, 12, and 72 keypoints (a single 3D keypoint, a limb, and the full body, respectively). **(A)** Trained hierarchical model (HBMAE, solid) versus its embeddings after average pooling (HBMAE Pooled, dashed). Action and activity F1 remain largely stable across pooling levels, whereas moveme F1 drops substantially, mirroring the trend seen in the trained layers and indicating that the per-level moveme decrease is driven in large part by spatio-temporal resolution, while the action/activity gains are not. **(B)** Multi-scale decoding (HBMAE Multi-Decoding, solid) versus the pre-fusion embeddings of each level extracted before the summation step and evaluated at the final layer’s resolution (HBMAE Multi-Decoding Fused, dashed). This isolates the learned pooling of the fusion head from the transformer blocks: the last layer matches the multi-scale result by construction, the first layer improves after pre-fusion, and moveme prediction is consistently reduced by the learned pooling.

fine-grained temporal structure is lost when representations are projected to the coarser spatial resolution of the last stage. Nevertheless, the learned pooling still outperforms simple average pooling overall, confirming that the reconstruction objective and the behavioural encoding objective share some common structure, even if imperfectly aligned.

5.1.2 Multi-Stage Decoding Strategy Does Not Improve Behavioural Encoding

Although the multi-scale decoding strategy does not improve linear probing performance on Shot7M2, it performs on par with single-scale decoding and allows the contribution of each encoder level to be studied independently. The layer ablation study across training epochs reveals two distinct learning phases (Fig. 5.3A). In early epochs, all single-scale conditions decrease uniformly from a shared starting MSE of 2.17×10^{-1} . As training progresses, the last layer (s2) drives further reconstruction improvement, with MSE values at convergence of 9.20×10^{-3} (only s0), 1.15×10^{-2} (only s1), 6.08×10^{-3} (only s2), and 1.10×10^{-3} for the full baseline. The fact that all single-scale MSE values remain higher than the combined baseline confirms that each level encodes complementary reconstruction-relevant information.

The leave-one-out results reveal a possible redundancy in the s1 level: ablating s1 alone produces a reconstruction error (5.33×10^{-3}) much closer to the baseline than ablating s0 (1.52×10^{-2}) or s2 (7.64×10^{-3}), suggesting that s1’s contribution is largely recoverable from the other two levels (Fig. 5.3A). This is further supported by the CKA analysis, which shows that s1 is more similar to the other levels (CKA of 0.518 with s0 and 0.777 with s2) than s0 and s2 are to each other (0.380), and by the magnitude analysis, where s1 has a notably smaller mean absolute value (2.33) compared to s0 (10.56) and s2 (8.51), suggesting a learned downscaling consistent with its relatively minor and redun-

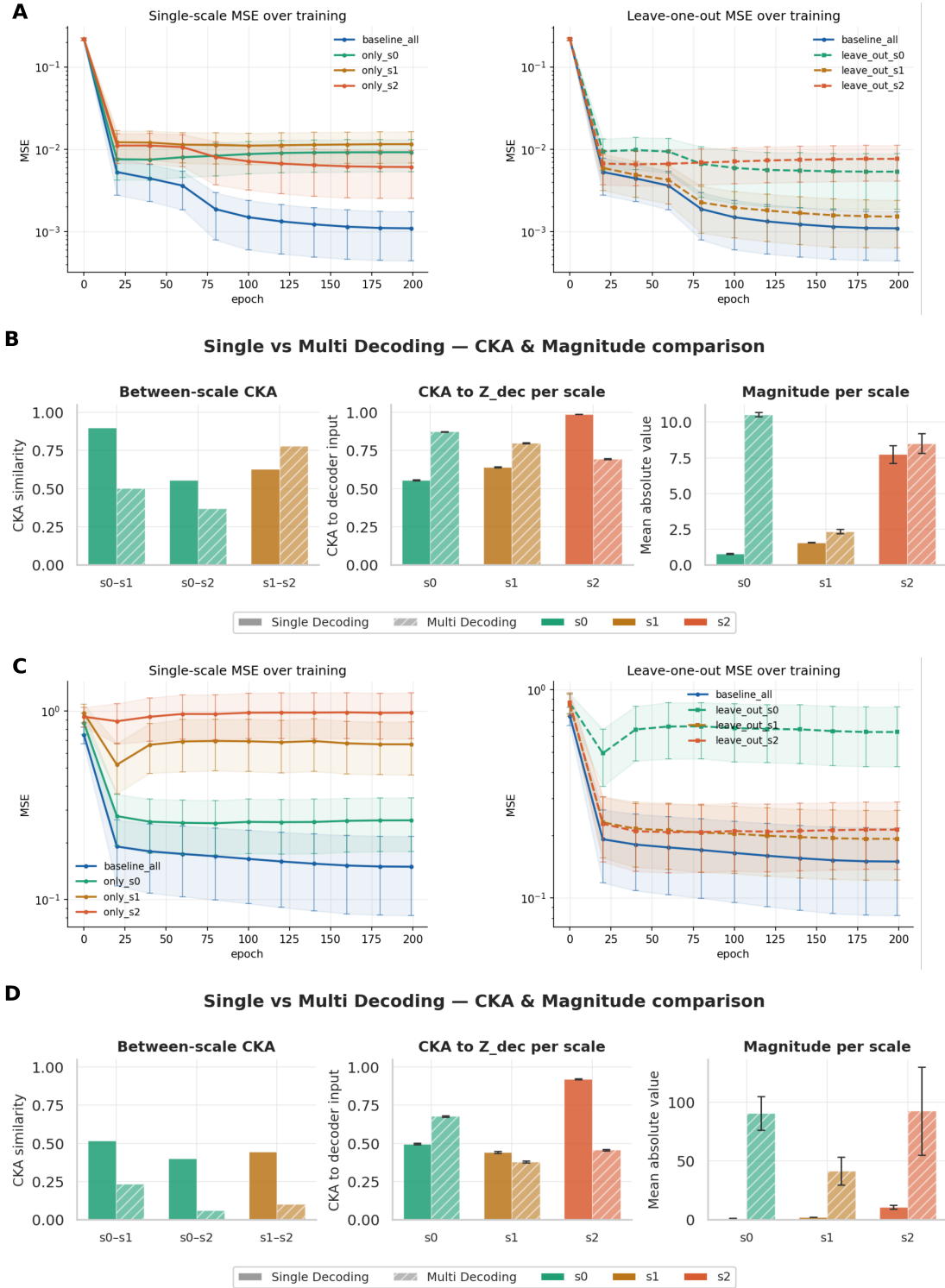


Figure 5.3: Layer ablation and representational analysis of the multi-scale decoding model, for Shot7M2 (A, B) and MABe22 (C, D). Throughout, s0, s1, and s2 denote the first, middle, and last encoder levels. (A, C) Reconstruction MSE over training epochs. *Left*: single-scale decoding, in which the decoder is given only one level’s embedding (only_s0/s1/s2) or all levels jointly (baseline_all). *Right*: leave-one-out decoding, in which a single level is ablated (leave_out_s0/s1/s2). On Shot7M2 (A) the last level (s2) drives most of the late-training improvement and all single-scale errors stay above the combined baseline, whereas on MABe22 (C) reconstruction relies predominantly on the first level (s0) and the single-scale curves remain essentially flat as the baseline continues to fall. (B, D) Single- versus multi-scale decoding (solid vs hatched bars) for Shot7M2 (B) and MABe22 (D), with colour denoting encoder level (s0, s1, s2). *Left*: between-scale CKA similarity (s0-s1, s0-s2, s1-s2). *Middle*: CKA between each level and the decoder input (Z_{dec}). *Right*: mean absolute embedding value per level.

dant reconstruction role (Fig. 5.3C). Returning to the linear probing results, s1 predicts all label types at a relatively consistent level across hierarchies without showing specialisation at any particular scale, which may reflect its role as a transitional representation between the fine-grained encoding of s0 and the behaviourally specialised encoding of the last layer. Importantly, the leave-out-s2 reconstruction error shows little improvement in later training epochs, while the baseline continues to decrease, indicating that the model primarily improves its last layer embeddings in the later stages of training. This observation directly motivates the nested dropout strategy and supports the assumption that s2 is the dominant contributor to the reconstruction objective.

The MABe22 ablation results offer a complementary explanation for the absence of hierarchical structure in that model. Here, the model learns to reconstruct almost entirely from the first layer, with single-scale MSE values at convergence of 0.263 (s0), 0.667 (s1), and 0.981 (s2), and a leave-one-out MSE for s0 (0.625) far higher than for s1 (0.192) or s2 (0.213) (Fig. 5.3B). The last layer shows virtually no improvement across training (MSE of 0.935 at epoch 0 vs. 0.981 at convergence), while s1 exhibits a transient improvement around epoch 20 (0.976 to 0.520) that does not persist at convergence (0.667). Strikingly, the single-scale MSE curves remain essentially flat throughout training, even as the full baseline MSE decreases to 0.149. This pattern suggests that the decoder learns to extract information from the existing embeddings without those embeddings themselves improving, a qualitatively different training dynamic from Shot7M2. The CKA values between layers of the MABe22 model are overall lower than in Shot7M2 (0.235, 0.053, 0.094 for s0–s1, s0–s2, and s1–s2 respectively), indicating that the levels encode more distinct representations, yet this diversity does not translate into hierarchical behavioural structure (Fig. 5.3D).

Taken together, the ablation analysis shows that the multi-scale decoding model trained on Shot7M2 primarily relies on the last encoder level for reconstruction, while the MABe22 model relies predominantly on the first. This asymmetry on one hand, might explain the discrepancy in the hierarchical structure obtained in the embeddings and on the other hand motivates applying nested dropout specifically to the Shot7M2 setting, where increasing the reconstruction difficulty of the last layer’s objective may encourage more informative use of the earlier levels.

5.1.3 Nested Dropout Sharpens Inter-Level Representational Distinctions

To properly evaluate the nested dropout implementation, four additional models were trained on Shot7M2 and compared via linear probing. For each of the hBehaveMAE and BehaveMAE baselines from Stoffl et al., a nested dropout variant was trained with a minimum keep rate of 0.1 and exponential interpolation, resulting in a keep rate of 0.1 for the last layer and 1.0 for the first. A high-mask comparison model with masking ratio 0.95, corresponding to the cumulative effect of the base masking ratio (0.70) combined with a 0.1 last-layer dropout, was also trained to disentangle the effect of increased masking difficulty from the directional nature of nested dropout.

Nested dropout induces a clear inter-level specialisation even in the non-hierarchical BehaveMAE model, where no spatio-temporal pooling constraints are imposed. The last layer shifts toward high activity prediction (0.787) and low moveme prediction (0.468), while the first layer shows the opposite pattern (activity 0.716, moveme 0.630), compared to the non-hierarchical baseline where both layers remain largely undifferentiated (last layer: 0.739, 0.623; first layer: 0.718, 0.643) (Fig. 5.4A,B). When compared to the high-

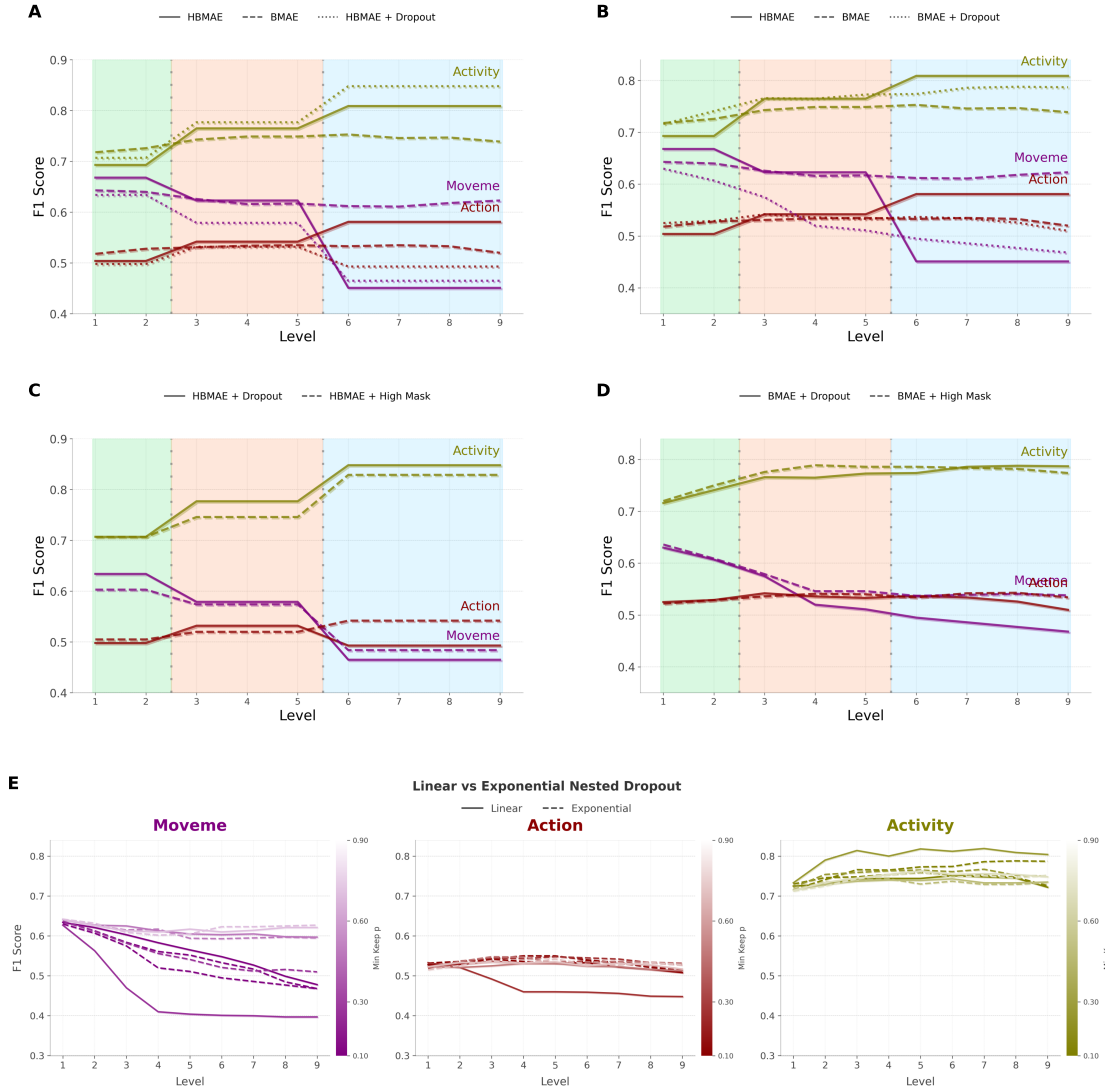


Figure 5.4: (A–D) Best-layer F1 score per encoder level for movemes (purple), actions (red), and activities (olive); background shading denotes the three hierarchical blocks. (A) Hierarchical model with nested dropout (HBMAE + Dropout, dotted) compared with the hBehaveMAE (HBMAE, solid) and BehaveMAE (BMAE, dashed) baselines. (B) Non-hierarchical model with nested dropout (BMAE + Dropout, dotted) compared with the same two baselines. (C) Hierarchical nested dropout (HBMAE + Dropout, solid) versus a high-mask control (HBMAE + High Mask, dashed; masking ratio 0.95), isolating the directional effect of the dropout from the increased masking difficulty. (D) The same comparison for the non-hierarchical model (BMAE + Dropout vs. BMAE + High Mask). (E) Minimum-keep-rate sweep of the nested dropout schedule, shown separately for movemes, actions, and activities. Line style distinguishes linear (solid) from exponential (dashed) interpolation of the per-level keep rate, and colour encodes the minimum keep rate (Min Keep p , 0.1–0.9). Very high keep rates recover the undifferentiated multi-decoding behaviour, whereas a minimum keep rate near 0.1 produces the strongest specialisation without degrading activity prediction.

mask model, however, this improvement is less pronounced (last layer: 0.774 activity, 0.532 moveme; first layer: 0.720 activity, 0.636 moveme), indicating that a large part of the effect in the non-hierarchical setting can be attributed to the increased overall masking difficulty rather than to the directional structure of the dropout.

This is not the case for the hierarchical hBehaveMAE model. Nested dropout increases activity prediction in the last layer from 0.809 (single-decoding baseline) to 0.848, and this improvement persists when compared to the high-mask counterpart (0.829), confirming that the combination of hierarchical pooling and directional dropout produces a genuine additional benefit (Fig. 5.4C,D). Furthermore, the directional structure of the dropout, which applies lower dropout rates to earlier layers, appears to preserve fine-grained temporal information in those levels, improving moveme prediction in the first layer to 0.634 compared to 0.603 in the high-mask model.

The min-keep rate sweep confirms that exponential interpolation produces smoother per-level keep rate curves, as expected, and that very high keep rates recover the undifferentiated behaviour of the standard multi-decoding model (Fig. 5.4E). Within the range explored, a min-keep rate of 0.1 appears close to optimal, with no observed degradation of activity prediction at this dropout level. Keep rates above 0.5 produce negligible differences relative to the no-dropout baseline, suggesting that only sufficiently aggressive dropout of the last layer drives the observed specialisation.

5.2 Arena

5.2.1 Positional Information Carries Limited Predictive Power for Behavioural Labels

To justify the choice of centre-alignment and the removal of positional information from the input, four models were trained with different input configurations: Centre-Align (A), No Centre-Align (B), Centre-Align without positional information (C), and positional information only (D). Per-frame linear probing was used to evaluate both the label predictability and the presence of any hierarchical structure across encoder levels, following the protocol of Stoffl et al. and including the alternate syllable labels obtained from the Keypoint-MoSeq analysis as additional per-frame targets (Fig. 5.5A).

Config C achieves the best strain prediction with a macro F1 of 22.9, while positional-only config D performs on par for experimental phase (42.0 vs. 40.9 for config C) and age prediction (57.8 vs. 60.8 for config C), indicating that absolute arena position is as informative as keypoints for phase and age but significantly less for strain identity. Centre alignment consistently improves over the normalised raw keypoints of config B across most labels (strain: 14.2 vs. 9.2; experimental phase: 41.0 vs. 43.7; age: 59.2 vs. 59.1), with a particularly pronounced gain in syllable prediction (12.2 vs. 6.7), suggesting that separating head direction, global position and keypoints information substantially clarifies the body-pose signal for behavioural analysis (Fig. 5.5A). The syllable prediction gain with centre-alignment is expected, as the Keypoint-MoSeq model that generated these labels was itself trained on centre-aligned data.

Regarding hierarchical structure, most label prediction scores increase monotonically with encoder level depth, sometimes peaking at an intermediate layer, with no clear decrease in higher levels, including for the syllable labels. This uniform pattern across all prediction tasks indicates an absence of level-specific specialisation in the Arena embeddings, consistent with the findings on MABe22. The encoding of behavioural information appears to increase gradually or remain stable across levels regardless of the label type,

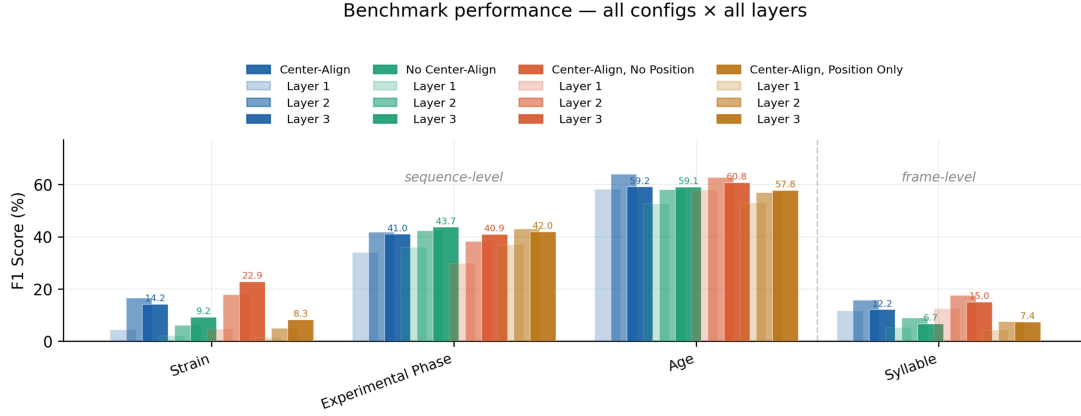
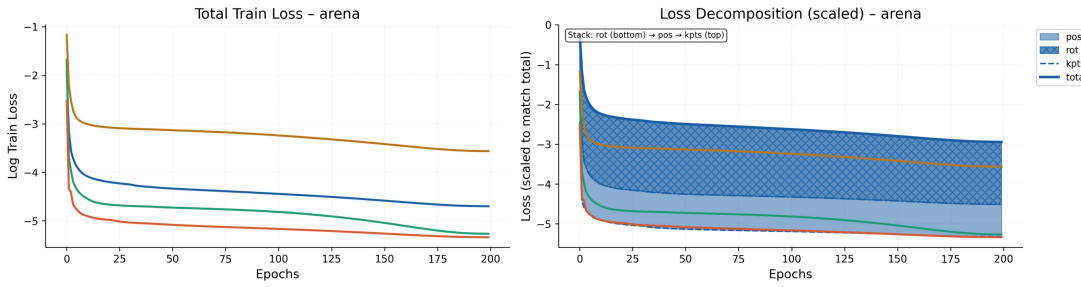
A**B**

Figure 5.5: Four configurations were compared: Centre-Align (blue), No Centre-Align (green), Centre-Align without position (red), and Centre-Align position-only (gold). **(A)** Macro F1 score (%) per label for each configuration, broken down by encoder level (Layer 1–3, light to dark within each colour). The targets comprise three sequence-level labels (strain, experimental phase, age) and one frame-level label (syllable, from the Keypoint-MoSeq analysis); selected per-bar values are annotated. **(B)** Training dynamics on the Arena dataset. *Left*: total training loss (log scale) over epochs for each configuration. *Right*: decomposition of the loss into its keypoint (kpts, dashed), position (pos), and rotation (rot) components, scaled to match the total (solid); components are stacked rot (bottom) → pos → kpts (top).

rather than exhibiting the coarse-to-fine specialisation observed on Shot7M2.

The training dynamics provide additional evidence in favour of config C (Fig. 5.5B). Notably, jointly training on keypoints and positional information does not improve the keypoint reconstruction loss, contradicting the assumption that positional information might carry keypoint-correlated signals, for example, a mouse turning more at the periphery of the arena or moving more linearly in the centre. Either this co-variation is absent in the data or the model is unable to exploit it under the current training setup. Finally, all centre-aligned configs show a plateau in training loss that is not observed in config B. This suggests that the no-centre-align model could potentially reach comparable classifier performance given longer training. In fact, centre alignment seems to substantially accelerate convergence by decoupling body-pose information from arena-position information, thereby simplifying the model’s reconstruction objective.

5.2.2 Sequence-Level Embedding Aggregation Maximises Label Predictability

To evaluate the length-agnostic implementation of hBehaveMAE, an additional model was trained with identical parameters to the centre-aligned, no-position model described

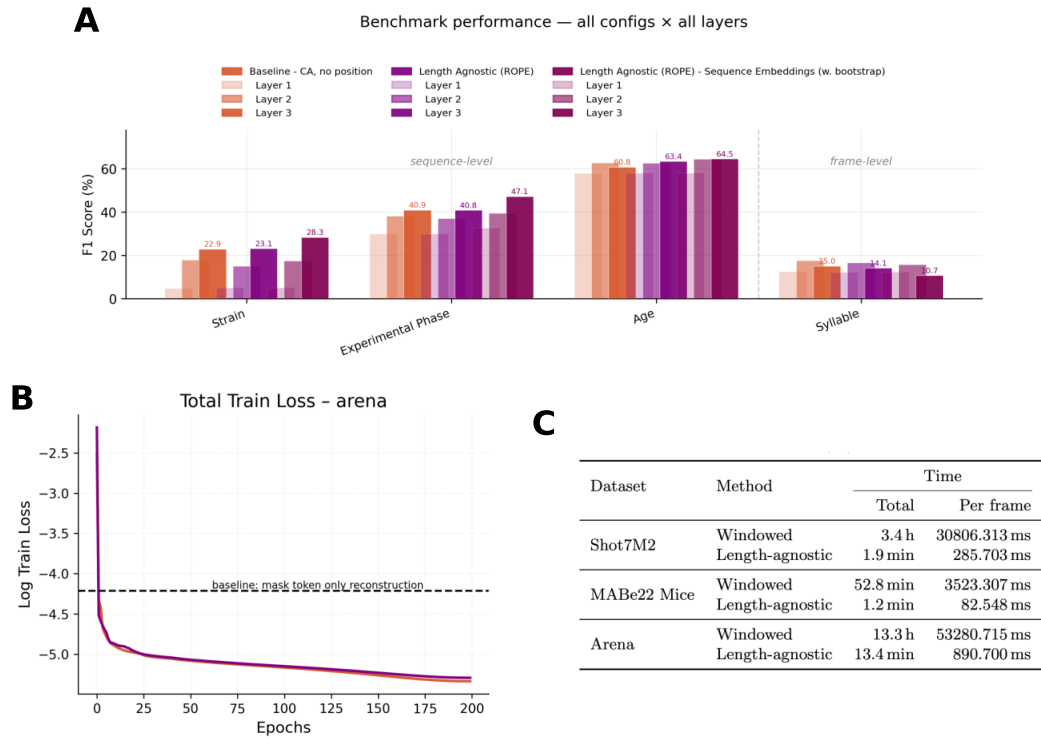


Figure 5.6: **(A)** Macro F1 score (%) per label for three configurations — learned-embedding baseline (orange), RoPE (purple), and RoPE with full-sequence extraction (dark red) — broken down by encoder level (Layer 1–3, light to dark). Targets: three sequence-level labels (strain, phase, age) and one frame-level label (syllable); selected values annotated. **(B)** Total training loss (log scale) over epochs; the dashed line marks the mask-token-only baseline. RoPE leaves the reconstruction dynamics essentially unchanged, and the gains from full-sequence extraction are not accompanied by a lower reconstruction loss. **(C)** Wall-clock extraction time, windowed versus full-sequence length-agnostic, across all three datasets (single NVIDIA H200 GPU, EPFL RCP cluster). The varying per-frame time reflects the spatial resolution of the data and the model configuration.

above, differing only in its temporal positional embeddings: learned embeddings were replaced by RoPE. This substitution does not appear to hurt the quality of the obtained embeddings, with on-par or improved prediction scores across all labels compared to the learned embedding baseline: experimental phase F1 of 40.8 vs. 40.9, strain of 23.1 vs. 22.9, and age of 63.4 vs. 60.8, using the same windowed extraction method. This confirms that fixed RoPE encodings are at least as informative as learned temporal positional embeddings for this dataset. Additionally, the speed of extraction was assessed on Shot7M2, MABe22 and the Arena dataset, where it was shown to perform up to $108\times$ faster on the Shot7M2 dataset and reduced the extraction time from 13.3 hours for the windowed extraction on the Arena dataset to a quick 13.4 minutes (Fig. 5.6C).

More interestingly, the full-sequence extraction method, which requires only a single forward pass of the encoder per recording, produces richer embeddings than the windowed approach despite its greater simplicity. Strain F1 increases to 28.3, experimental phase to 47.1, and age to 64.5 (Fig. 5.6A). The only task where full-sequence extraction underperforms is alternate syllable label prediction, which drops from 15.0 with windowed extraction to 14.1 with length-agnostic windowed extraction and 10.7 with full-sequence extraction. This degradation is expected: syllable labels are defined at a fine temporal resolution, and the full-sequence extraction repeats the last-layer embedding over 10 frames,

producing a coarser and less precise temporal alignment with the per-frame syllable annotations. These embeddings were therefore adopted for all downstream experiments. Finally, the training loss curves confirm that RoPE embeddings do not affect the reconstruction dynamics relative to the learned embedding model, and that the embedding quality improvement obtained with full-sequence extraction is not explained by a lower reconstruction loss but could be explained by the additional contextual information captured through a full-sequence attention pass over the behavioural recording (Fig. 5.6B).

5.2.3 hBehaveMAE Embeddings Outperform Commonly Used Behavioural Representations

To evaluate the embeddings obtained from hBehaveMAE trained on the Arena dataset, several baselines were implemented. These include the raw pose representation, a hand-crafted behavioural feature representation (see Appendix Additional Methods), Keypoint-MoSeq applied to the centre-aligned keypoints and constrained to 50 syllables, and two deep learning baselines: TS2Vec and TCN. Since all Arena labels are per-sequence, sequence-level embeddings were obtained from all methods and evaluated using the per-sequence linear probing protocol.

hBehaveMAE outperforms all baselines on every task, achieving age F1 of 0.74, experimental phase F1 of 0.76, and a particularly strong strain prediction of 0.772, compared to the second-best method TS2Vec at 0.40 (Fig. 5.7A). Keypoint-MoSeq and TS2Vec both achieve reasonable results (age/phase/strain of 0.64/0.49/0.32 and 0.70/0.65/0.40 respectively) but do not consistently improve over the simple behavioural feature baseline (0.61/0.52/0.34), suggesting that the additional representational capacity of these methods does not translate to a net gain on this dataset. Crucially, applying k -means clustering to the hBehaveMAE embeddings with $k = 50$, matching the number of Keypoint-MoSeq syllables, still outperforms Keypoint-MoSeq directly (0.70/0.61/0.58), demonstrating that the performance advantage of hBehaveMAE is not simply a consequence of the continuous, high-dimensional nature of its embedding space but reflects a genuine improvement in the organisation of the behavioural space itself.

To further characterise this behavioural space, its alignment with the genomic space was assessed using a Mantel test with Spearman correlation. Genetic distances were computed from SNP data using two panels: CGD-MDA1 (30 overlapping strains, 470,822 SNPs) and CGD-MDA2 (28 strains, 132,261 SNPs). The hBehaveMAE embedding space shows a stronger correlation with genetic distance than Keypoint-MoSeq across both panels: Mantel r of 0.071 and 0.188 for the hBehaveMAE cluster embeddings, compared to 0.030 and 0.119 for Keypoint-MoSeq. The only statistically significant correlation is observed on the CGD-MDA2 panel for the hBehaveMAE embedding space ($r = 0.255$, $p < 0.001$) (Fig. 5.7C,D). Interestingly, CGD-MDA2, the panel yielding the strongest genotype–phenotype correlation, contains the strains with the lowest classification F1 scores, which at first appears paradoxical. This either suggests a weaker linear relationship between genetic distance and behavioural discriminability among these strains, or reflects a fundamental difference between the two evaluation protocols. Linear probing captures the information accessible to a linear classifier, which is governed by the separability of class centroids; it does not directly reflect the spatial organisation of the embedding space as a whole. The Silhouette score would be a more appropriate measure of the latter and is discussed in the following section.

Finally, it is worth noting the large discrepancy between the strain F1 score obtained with the per-sequence probe (0.772) and the per-frame pipeline discussed above (0.283).

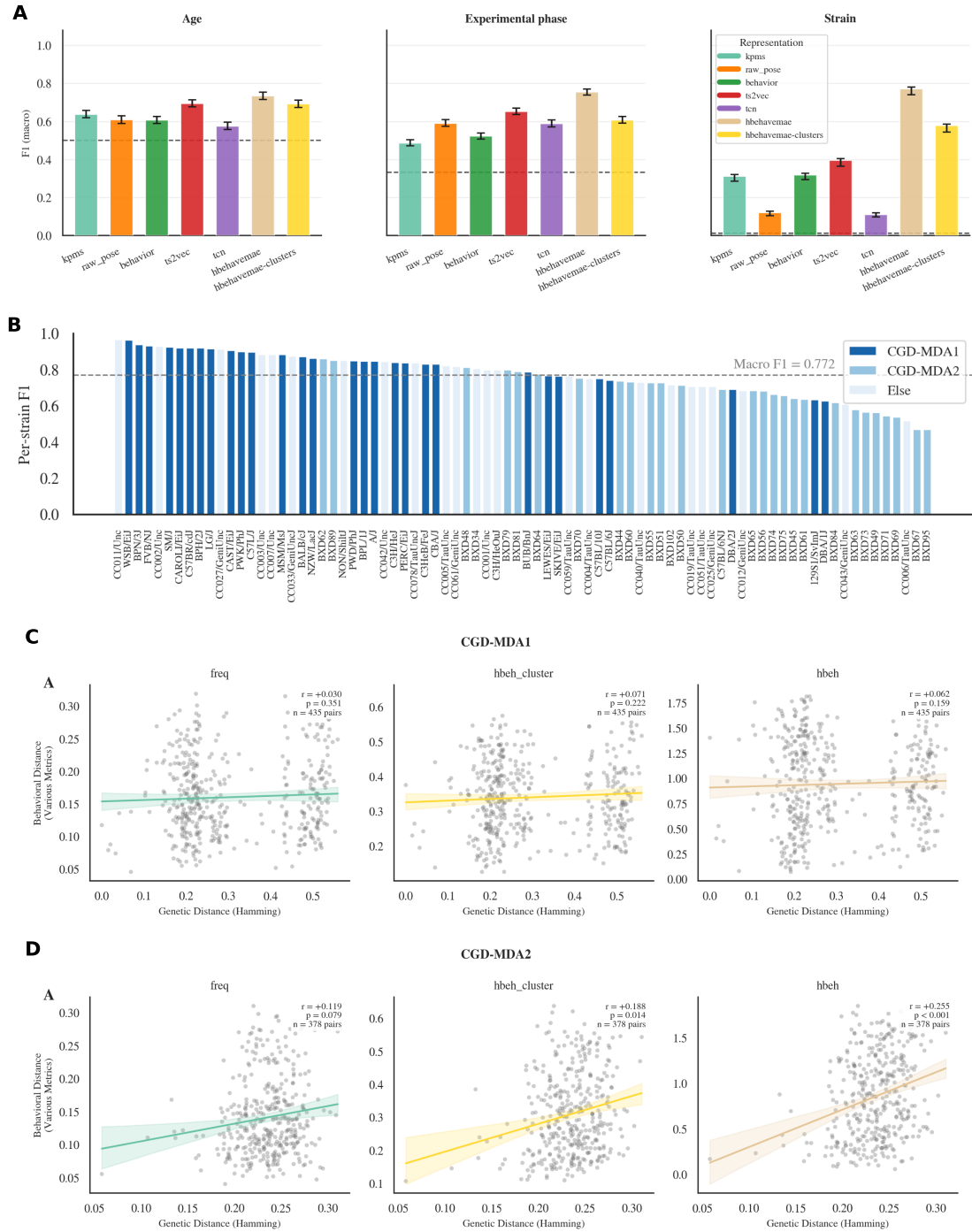


Figure 5.7: **(A)** Per-sequence linear-probing F1 score for each representation — Keypoint-MoSeq (kpms), raw pose (raw_pose), hand-crafted behavioural features (behaviour), TS2Vec (ts2vec), TCN (tcn), hBehaveMAE (hbehavemae), and k -means clustering of the hBehaveMAE embeddings with $k = 50$ (hbehavemae-clusters) — on three sequence-level tasks: age, experimental phase, and strain. **(B)** Per-strain F1 score for hBehaveMAE on the strain task, sorted in descending order and coloured by SNP-panel membership (CGD-MDA1, CGD-MDA2, or neither); the dashed line marks the macro F1 of 0.772. **(C, D)** Mantel-test relationship between behavioural distance and genetic (Hamming) distance for the CGD-MDA1 (C, $n = 455$ strain pairs) and CGD-MDA2 (D, $n = 378$ strain pairs) SNP panels, each shown for three representations: Keypoint-MoSeq syllable frequencies (freq), clustered hBehaveMAE embeddings (hbeh_cluster), and hBehaveMAE embeddings (hbeh). Points are strain pairs, lines show the linear fit, and the Mantel r , p -value, and pair count are annotated in each panel.

This difference stems directly from the training procedure: the per-frame classifier sees only one frame at a time and, although its predictions are subsequently averaged across frames, it never has access to the full sequence context during training. Averaging the per-frame embeddings across an entire recording before training the classifier, as done in the per-sequence pipeline, provides a much richer and more stable representation of each animal’s behavioural phenotype, leading to substantially better classification across all tasks.

5.2.4 Age and Experimental Phase Structure the Continuous Behavioural Manifold

The ageing trajectory of a strain can be characterised in the behavioural embedding space by computing the vector connecting the mean adult embedding to the mean old embedding. This displacement captures the direction and magnitude of behavioural change due to ageing for a given strain and experimental phase. To focus on strains with the most reproducible ageing trajectories, the five strains with the highest bootstrapped cosine similarity across age groups were selected: DBA/2J (0.68), BPN/3J (0.70), BPH/2J (0.77), BXD89 (0.87), and BXD34 (0.89) (Fig. 5.8A,B). The consistency of each strain’s trajectory across experimental phases was also assessed, defined as the cosine similarity between the habituation and retention phase embedding vectors. BPH/2J shows the highest phase-trajectory consistency among the selected strains (0.93), followed by BXD89 (0.72), BPN/3J (0.47), BXD34 (0.43), and DBA/2J (0.23) (Fig. 5.8C).

The t-SNE projection of the embedding space reveals clear strain-specific ageing structure, with each strain displaying a distinct ageing direction that varies across experimental phases (Fig. 5.8A). BPN/3J shows a particularly pronounced separation between habituation and the other phases, consistent with its being the strain with the most phase-consistent ageing trajectory. Since t-SNE does not preserve inter-point distances, all quantitative comparisons were performed directly in embedding space; the manifold projection is used for visualisation only.

BXD34 shows both the most consistent age trajectory and the largest mean adult-to-old L2 distance across phases (1.05, 0.92, and 0.60 for HAB, ACQ, and RET respectively), suggesting a substantial and reproducible behavioural shift with ageing (Fig. 5.8D). In contrast, BXD89 shows much smaller between-phase variation in ageing displacement (0.43, 0.48, 0.46), indicating a more uniform behavioural change across experimental contexts (Fig. 5.8E). Finally, within-strain embedding spread, a measure of behavioural consistency within a given strain, phase, and age group, was computed for all strains. Among the five selected strains, DBA/2J is the only one showing a consistent reduction in spread with ageing across all three experimental phases, with L2 distances decreasing from 0.37 to 0.27, 0.64 to 0.30, and 0.58 to 0.44 for HAB, ACQ, and RET respectively, suggesting that older DBA/2J individuals express a more stereotyped and less variable behavioural repertoire (Fig. 5.8F).

The consistency of the results across the five selected strains supports the validity of the hBehaveMAE behavioural space as a meaningful representation of individual and strain-level behavioural phenotypes. A current limitation of this analysis, however, is the difficulty of distinguishing whether a high embedding spread for a given strain reflects genuinely complex and variable behaviour, or a weaker embedding representation for that strain. Two strains could in principle be behaviourally indistinguishable regardless of the analysis method used, and the current pipeline does not provide a means to adjudicate between these interpretations.

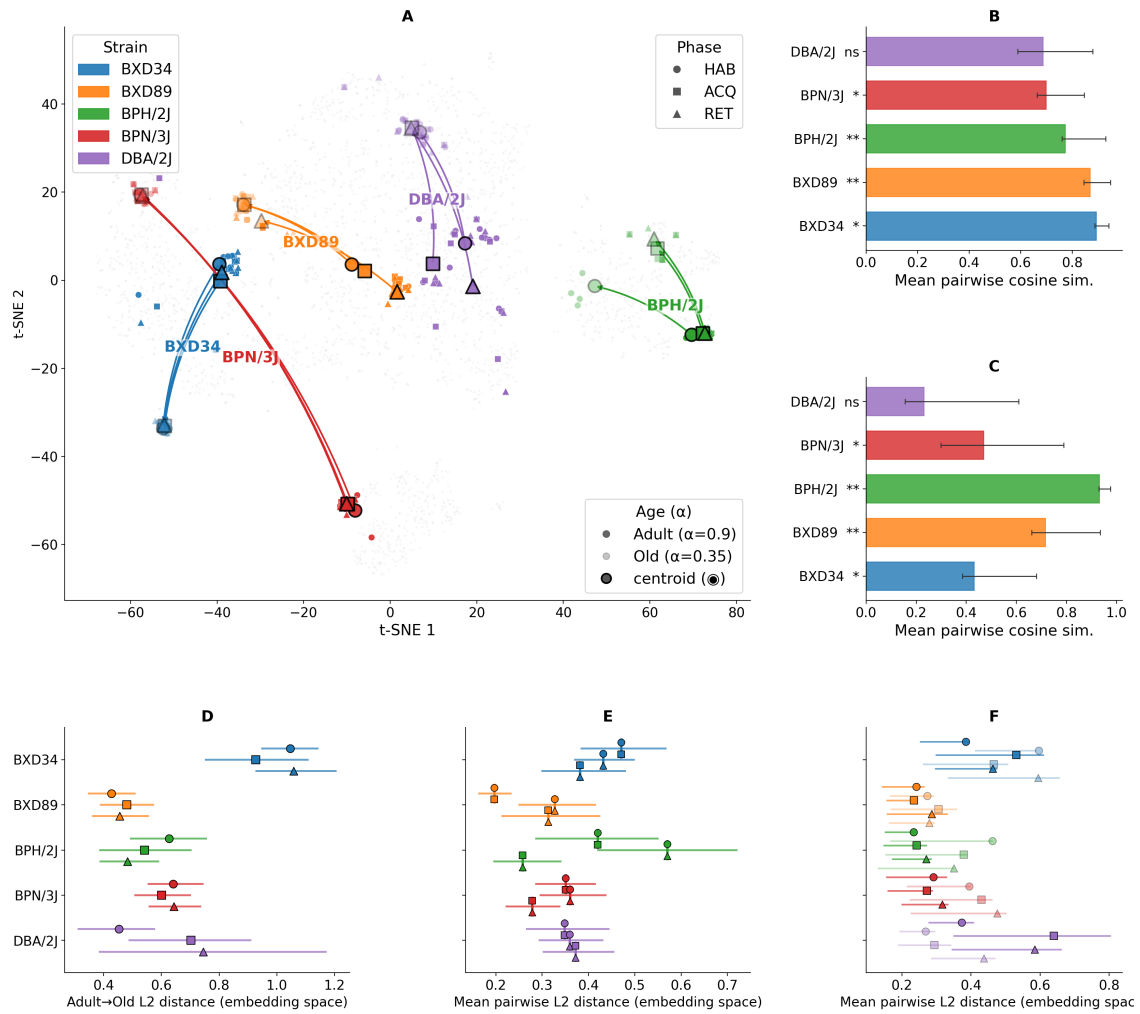


Figure 5.8: Age- and phase-resolved structure of the hBehaveMAE behavioural manifold for the five selected strains (BXD34, BXD89, BPH/2J, BPN/3J, DBA/2J). Throughout, marker shape encodes experimental phase (HAB: circle, ACQ: square, RET: triangle) and, where shown, opacity encodes age group (adult: saturated; old: faded). **(A)** t-SNE projection of the embedding space. Small markers are individual recordings (grey: full dataset), large outlined markers are strain $\times$ phase centroids, and curved arrows connect each strain's adult and old centroids, depicting its ageing trajectory per phase. t-SNE is used for visualisation only; all quantitative comparisons were performed directly in the embedding space. **(B)** Mean pairwise cosine similarity of the ageing direction across age groups, per strain (the strain-selection criterion); higher values indicate more reproducible ageing trajectories. **(C)** Phase-trajectory consistency, quantified as the cosine similarity between the habituation and retention ageing vectors, per strain. **(D)** Adult-to-old L2 distance (magnitude of the ageing displacement) per strain, separated by phase. **(E)** Within-strain mean pairwise L2 distance (embedding spread) per strain, separated by phase. **(F)** Within-strain mean pairwise L2 distance per strain, separated by phase and age group (adult: saturated, old: faded), showing the change in behavioural spread with ageing. Error bars indicate 95% bootstrap confidence intervals (percentile method, 2000 resamples); significance annotations in (B, C) denote one-sided permutation test against a pooled null (1000 permutations), Benjamini–Hochberg FDR-corrected across strains: ns, not significant; * and **, increasing significance.

Chapter 6

Discussion

A recurring tension in behavioural analysis is that the methods used to uncover the structure of behaviour often impose that structure themselves. This thesis built on hBehaveMAE to ask how far an unsupervised, reconstruction-based model can reduce these assumptions.

6.1 What drives hierarchical representation learning

Studying the behavioural space obtained from hBehaveMAE across different datasets allowed us to outline the impact of the hierarchical structure of the data modality on the resulting latent space. The synthetic nature of Shot7M2 was shown to lead to a more hierarchically structured embedding space than the more complex natural movements of MABe22 and the Arena dataset, where the hierarchy of behaviour is less pronounced. The three datasets differ along two axes that jointly explain this gap: the strength of the raw hierarchical signal in the data, and the degree to which the available labels describe it. Shot7M2 scores high on both: it produces a strong raw hierarchical signal at different temporal resolutions, and it provides hierarchically nested, informative labels. These two aspects combined greatly simplify the classification task. MABe22, on the other hand, has labels that were shown to be much less informative about the data, and that are spatio-temporally much less correlated with the structure fixed by the pooling kernels of the trained hBehaveMAE models. The Arena dataset, finally, lacks any hierarchical labelling at all which would describe such a hierarchy. This observation shows the difficulty of defining model hyperparameters when studying natural hierarchy. In fact, when the hierarchy is not as well spatio-temporally defined as in the Shot7M2 dataset, which hierarchical structure should be adopted, and thus which pooling stages? Interestingly, while the spatial structure fixed by the kernel of the Shot7M2 model matches closely the labels (from limbs to full body in the last stage), the temporal structure doesn't match at all with a maximum of 8 frames in the final stage for a median length of action labels of 25 and 1800 frames for activity. As discussed in the method section the masking strategy wouldn't allow pooling up to 1800 frames this observation was also a driving choice towards a maximum of 10 frames temporal pooling for the Arena dataset.

Additionally, layer ablation showed a significant difference in the learning process of Shot7M2 and MABe22. The first model, trained on Shot7M2, learned to use its last layer for reconstruction while the latter mostly relied on its first layer, effectively not learning any relevant hierarchy from the structure imposed by the pooling stages. Notably, the second layer of the Shot7M2 hierarchical model was shown to be somewhat redundant with respect to its higher and lower hierarchical layers, possibly attesting to too high a number

of layers. This echoes an observation discussed in the background section regarding the difficulty of defining the number of relevant hierarchical layers for a given behaviour, and reinforces this thesis’s argument for reducing structural assumptions within the model.

6.2 The nature of the recovered hierarchy

Further studying the embeddings obtained from the Arena dataset, an unexpected form of hierarchy arises. Although behaviour is widely regarded as hierarchically organised, and the Arena dataset should therefore hold some form of hierarchical structure, our labels are non-hierarchical, both temporally, with sequence-level labels, and spatially, with full-body labels. The hierarchy instead arises in the nested spatial organisation of the behavioural space. Since the embedding space of hBehaveMAE is generated by a reconstruction objective, distances within it can be interpreted as differences in raw pose. Clusters in this space therefore represent behaviourally similar sequences to some degree. These clusters were shown to be organised hierarchically: first distinguishing clearly between different strains or the age of an animal, and then, in a nested fashion, encoding differences in behaviour related to the experimental paradigm of the sequence. We observe that strain and age are not inherently nested: some aged subpopulations lie further from other strains than from their adult counterparts, and the ageing trajectory does not appear to be shared across strains, with each strain seemingly following its own ageing process. This suggests that the experimental paradigm explains less variance in behaviour than either age or strain. Interestingly, BXD34 shows the largest Adult→Old distance in the embedding space, suggesting that its behaviour undergoes the most pronounced change with age among the strains examined. This stands in contrast to the neuroanatomical findings of Johnson et al. [18], who identified BXD34 as the strain with the smallest whole-brain connectivity change with age across 13 BXD strains, including BXD89. Whether this discrepancy reflects a genuine difference between structural and behavioural ageing trajectories, or emerges from confounds such as morphological differences captured by the pose estimator, remains an open question.

Interestingly, and similarly to the shallow brain hypothesis, this hierarchy emerges from a single continuous space. It is thus not spatio-temporal in nature but rather parallel to it, and closer to the hierarchy defined by FlexTok: a hierarchy in terms of amount of information or level of abstraction. In this idea, coarser representations capture the broad category of an object before finer details are added progressively. In our case, the coarsest abstraction corresponds to the behaviour of a strain at a specific age, and the experimental design provides finer information about behaviour within a sequence. It would therefore be interesting to study whether specific dimensions encode particular levels of this hierarchy, and how they correlate with the variance explained at each level. This process would ultimately help clarifying which factor most strongly shapes behaviour: strain, age, or experimental phase. It would also be worth investigating whether this hierarchical organisation still emerges from the hierarchical pooling stages of the model, or simply as a result of the label-related variance present in the dataset. This could be tested by training a non-hierarchical BehaveMAE model with the same configuration used here.

Importantly, the hierarchical organisation observed, combined with the high strain prediction accuracy obtained is a possible mark of leakage of body-shape information. The attention mechanism of the transformer may pick up subtle morphological differences unrelated to behaviour. This bias could significantly structure the embedding space and allow a linear probe to easily classify strains. Testing this hypothesis would require a

dedicated analysis of morphological variance in the pose estimates and its correlation with both age and strain. Additionally, a morphology-only probe could be trained only on inter-individual morphological differences to obtain a baseline to compare to, although the raw pose baseline already evaluated in per-sequence linear probing is, to some extent, a safeguard against strong pose-related biases.

6.3 hBehaveMAE as a behavioural representation

When comparing the behavioural space obtained from hBehaveMAE to that of other methods, hBehaveMAE was shown to outperform competing approaches on both linear probing and genetic distance correlation. One caveat is that TS2Vec was trained on a subset of the data rather than the full dataset (see methods); its results here are therefore a lower bound on its potential performance. Notably, hBehaveMAE outperforms all other methods even when the continuous embedding space is discretised into 50 clusters via K-means. This interesting finding supports the view that behaviour is intrinsically continuous and that discretisation is only a lossy simplification of it. Additionally, the marked difference between per-frame and per-sequence probing results tells us more about the organisation of the behavioural space: while fine-grained frame-level differences are harder to detect with a linear model, averaging embeddings over a full sequence makes the classification task considerably easier. The length-agnostic extraction strategy naturally exploits this, with attention benefiting from the behavioural context of the full sequence.

A discrepancy remains between the strong linear probing results and the low genetic distance correlations. While predicting 80 different strains with an F1 score of 0.77, no or very little significance is observed between the retrieved behavioural space and the computed genetic space. This observation, however, cannot tell us whether this discrepancy is due to a genuinely weak genotype-to-phenotype link within those strains or whether the behavioural space does not capture the relevant behavioural differences. Notably, we observed that the strains predicted with the greatest accuracy in linear probing were not the strains whose genetic background correlated most with the obtained behavioural space. This could suggest that the learned behavioural difference between strains is uncorrelated to their genetic distance.

Importantly, the behaviour studied in this thesis need not be taken out of its context. Although open-field activity is often regarded as a sign of complex natural behaviour, the Arena dataset is strongly populated by wall-hugging and escape-finding, as the first figure of this thesis, in the abstract, shows. While open-field behaviour has become a baseline of so-called natural behaviour in experimental sessions, we should be careful when defining it as such, since it is actually an experimental paradigm in itself, that of a captured animal. On this aspect, several studies have tried to extrapolate to the study of true environments, such as the MammAlps [11] dataset, which studies animal behaviour non-invasively in its natural setting. Such "true" natural behaviour, however, does not allow experimental manipulations, or renders them very difficult, and the open-field paradigm remains very useful.

6.4 Architectural insights and design trade-offs

Although still at a proof-of-concept stage, the technical contributions of this thesis show promising results. The length-agnostic implementation allowed for variable-length embedding extraction, which showed improved performance in linear probing while also being much more efficient than the past windowed extraction. Testing this implementation on

Shot7M2 and MABe22 was not deemed relevant, as these datasets are of fixed length. However, it would be interesting to study whether this improvement in embeddings remains across configurations, pooling stages, and embedding dimensions. In addition, a slight decrease in performance was observed in the alternative frame-labels retrieved from Keypoint-MoSeq syllables. This possible lack of fine resolution, as a result of the full-sequence extraction, should be further studied in a true hierarchical variable-length dataset. The potential of the length-agnostic implementation was, however, not fully used, as the training scheme was kept to a fixed window. In fact, as discussed in the supplementary section, variable-length inference also allows for variable-length training, which is often used in LLM or multimodal training. The results were not conclusive enough yet and would demand increased research into the masking strategy, and into whether to keep it constant across variable lengths or to make it vary.

Nested dropout showed promising results on Shot7M2, where it both allowed a form of hierarchically organised embedding layers to be retrieved without any spatio-temporal pooling in the non-hierarchical model, and increased the difference between the early and later hierarchical layers' linear probe specialisations in the hierarchical model. These results are, however, very limited: in the non-hierarchical model the effect was shown to be largely a consequence of the higher masking ratio, whereas in the hierarchical model it persisted against the high-mask comparison. The nested aspect of the dropout was additionally not specifically studied and should thus be separately ablated to study its relevance. Most importantly, nested dropout could completely replace the MAE masking strategy. In fact, in this study it was only applied to at most 30% of the input, while the rest was still uniformly masked. Ridding it completely of this uniform mask could increase the potential effect of the nested dropout. However, this would also lead to a considerable loss in efficiency and a redesign of the loss computation, so far only computed on masked tokens. Additionally, allowing all tokens of the finest layer to be visible could make the reconstruction task too easy, and thus careful pooling strides and patch kernels should be chosen to ensure a small enough bottleneck.

Several other approaches were also studied during this thesis but did not produce results promising enough to be deeply discussed. Considering the multi-decoding strategy of hBehaveMAE, one aspect seems, however, crucial to think about: at which resolution to reconstruct from. In fact, it was decided that reconstruction should occur at the coarsest definition, adding minimal changes from the single-decoding strategy and effectively reconstructing from the coarsest spatio-temporal resolution, but with additional information from earlier layers. This decision greatly limits the amount of pooling allowed before completely closing the bottleneck. While TS2Vec can build as many hierarchical representations as wanted without impacting the lower layers, all layers here are impacted by the resolution of the last layer of the model, which dictates the reconstruction task. If we were interested in obtaining a sequence-level representation, we would need to reconstruct from this spatio-temporal resolution and thus potentially mask any finer semantics learned in earlier layers. From this observation, upscale fusion was developed in opposition to the downscale fusion currently used. This method simply changes the fusion head to learn a linear transformation of later layers to match the spatio-temporal resolution of the first layer. This could also be further studied and potentially allow for more pooling stages.

However, all these modifications do not address the main limitation of the current hBehaveMAE architecture: the mask unit masking scheme. By masking at the coarsest level, we mask much more consecutive information than a simple random masking strategy would, as discussed through our mean loss of contextual information (MLCI) metric. When pooling a lot, we consequently mask a lot of consecutive information, effectively

losing finer temporal resolution. This masking scheme was rightfully chosen in order to not pull from masked tokens and to attend only to visible tokens. However, when applying nested masking in the fusion head, this masking strategy could be completely removed without ever hurting attention or the pooling kernels. This would potentially make upscale fusion much more efficient as well, by allowing tokens to be masked or dropped at the finer scale instead of masking the complete mask unit. This solution would, however, require deeper structural changes, which were out of the scope of this thesis.

6.5 Synthesis

Taken together, these results offer a partial answer to the question posed at the beginning of the discussion: how far can an unsupervised, reconstruction-based model reduce the structural assumptions imposed on behaviour? The clearest finding is that the model recovers a strong, level-specific spatio-temporal hierarchy only when the data carries one, as in the synthetic Shot7M2 dataset; on natural behaviour, where this signal is weak, the architectural hierarchy imposed by the pooling stages is largely not exploited. This suggests that hierarchy should emerge from the data rather than be fixed a priori, and motivates the two technical contributions of this thesis, the length-agnostic implementation which removes the fixed temporal window, and nested dropout which induces an ordered hierarchy without relying on spatio-temporal pooling.

Most strikingly, the hierarchy that does emerge on natural behaviour is not the one the architecture was designed to capture. Instead of a spatio-temporal hierarchy, the Arena embedding space is organised along an abstraction hierarchy more closely related to the hierarchy defined by FlexTok. This hierarchy separates coarse factors such as strain and age and finer, paradigm-related differences. This observation points towards a model whose hierarchy emerges from the data and the reconstruction objective rather than from imposed pooling stages. Realising this fully would require addressing the mask unit masking scheme identified above, and disentangling genuine behavioural structure from possible morphological leakage; these remain the most important open questions left by this work.

Code Availability

The full implementation supporting this thesis, the nested-dropout and length-agnostic extensions to hBehaveMAE, all training and evaluation scripts, and the figure-generation pipeline, is available at <https://github.com/oskarboesch/MP-Repo.git>.

Supplementary Material

Supplementary methods and results are provided in the Appendices (Section Appendix), following the bibliography.

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Appendix

Use of Generative AI

The generative AI assistant Claude (Opus 4.8, Anthropic) was used to: assist in writing and debugging visualisation and figure-generation code, provide feedback on the organisation and structure of the manuscript and support language editing, including grammar and spelling correction.

All research design, model implementation, experiments, analyses, and the interpretation of results are the author’s own work. All AI-generated code and suggestions were reviewed, tested, and validated by the author, who takes full responsibility for the content of this thesis.

Additional Methods

Behavioral feature representations

All feature representations were derived from the same 8-point pose-estimation tracking (DeepLabCut), including the nose, head midpoint, left ear, right ear, neck, mouse centroid, mid-backend2, and tail base, tracked in 2D pixel coordinates at 25 fps. Keypoints with confidence below 0.3 were corrected by linear interpolation before feature extraction. Two complementary feature representations were derived from these raw keypoint trajectories and compared for downstream classification of age, experimental period, and strain.

Raw-pose statistical features. For each recording, every frame’s 8 keypoints (16 x/y coordinate values) were summarised across the full session using seven temporal statistics (mean, standard deviation, minimum, maximum, median, 10th percentile, 90th percentile), yielding 112 features per recording (16 coordinates $\times$ 7 statistics). This representation captures the raw distribution of keypoint positions without any behavioral interpretation.

Behavioural metrics. Thirteen behavioural metrics were computed per recording from the centroid trajectory (mean position across all keypoints) and from a three-point spine angle defined by the neck, mouse centroid, and tail base. Locomotion metrics included mean centroid speed, the variance of per-frame speed, mean and variance of frame-to-frame acceleration, the fraction of frames with speed below 25 px/s (stillness), and the fraction of frames with speed above 300 px/s (high-speed running). Space-use metrics included the fraction of time the centroid spent within a 180×180 px region centered on the arena (centre-time) and the proportion of total distance traveled while within this region (center-distance ratio). Posture metrics included the circular variance of body heading

(defined as the vector from neck to nose), the variance of frame-to-frame heading angular velocity, and the mean, mean angular velocity, and variance of the neck–centroid–tail-base joint angle.

Cluster Population Study

For each identified cluster, the median keypoint trajectory was computed by extracting all constituent 10-frame sequences, aligning them in egocentric space, and plotting the median position of each tracked body part across the sequence duration, providing a kinematic summary of the representative movement pattern. Spatial occupancy was visualised as a 2D positional heatmap over the circular open-field arena: for each spike assigned to a given cluster, the animal’s centroid position was recorded, and occupancy probability was estimated via kernel density estimation over the arena surface, yielding a non-linear probability map normalised to sum to one. Temporal usage profiles were computed by dividing each session into non-overlapping windows of 150 frames and calculating, for each window, the proportion of spikes belonging to a given cluster relative to all spikes in that window; the resulting time series were averaged across sessions and animals within each label group (age, experimental phase, or strain family), with shaded regions indicating ± 1 standard error of the mean across animals.

Embedding Space Distance Calculations

Strain-level trajectory analysis in the embedding space

To assess whether the hBehaveMAE embedding space captures biologically structured variation across age and experimental phase, a trajectory-based analysis was performed at the strain level. For each animal, a mean embedding was computed separately for each combination of age group (Adult, Old) and experimental phase (HAB, ACQ, RET), averaging over all recordings sharing those labels. Only animals with at least one recording in every cell of the 2×3 age $\times$ phase grid were retained for trajectory analysis, yielding 420 animals spanning 74 strains.

Ageing trajectory vectors For each animal, an ageing direction vector was defined as the difference between its Old and Adult mean embeddings, $\mathbf{v}_{\text{age}} = \bar{\mathbf{e}}_{\text{Old}} - \bar{\mathbf{e}}_{\text{Adult}}$, where each mean is taken across all phases. A phase-arc vector was defined as $\mathbf{v}_{\text{phase}} = \bar{\mathbf{e}}_{\text{RET}} - \bar{\mathbf{e}}_{\text{HAB}}$, where each mean is taken across both age groups.

Within-strain trajectory consistency The reproducibility of a strain’s ageing direction was quantified as the mean pairwise cosine similarity among all animals belonging to that strain:

$$\text{consistency}(\text{strain}) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{|\mathbf{v}_i| |\mathbf{v}_j|}$$

where n is the number of animals in the strain. The same metric was applied to phase-arc vectors to quantify phase-trajectory consistency. Only strains with at least 5 animals were considered. The five strains with the highest ageing-trajectory consistency were selected for detailed visualisation and further analysis.

Statistical testing Significance of the observed within-strain cosine similarity was evaluated against a one-sided permutation null distribution. For each strain, 1,000 permutation replicates were generated by drawing n animals uniformly at random from the full

pool of animals across all strains, computing the mean pairwise cosine similarity of their trajectory vectors, and recording the fraction of replicates exceeding the observed value. Raw p-values across the five strains were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure. Bootstrap 95% confidence intervals were estimated using the percentile method with 2,000 resamples (sampling animals with replacement within each strain).

Displacement and spread metrics The Adult-to-Old displacement magnitude was computed per animal and per phase as the L2 distance between the animal’s Adult and Old mean embeddings for that phase; the per-strain estimate is the mean across animals, with bootstrap confidence intervals computed as above. Between-phase distances (HAB–ACQ, HAB–RET, ACQ–RET) were computed as the mean pairwise Euclidean distance between the two sets of per-animal phase embeddings across all animals of a strain. Within-strain spread was quantified as the mean pairwise Euclidean distance among all session embeddings belonging to a given strain $\times$ phase $\times$ age cell, capturing the intra-strain variability in behavioural state under each condition.

Visualisation For visualisation purposes only, t-SNE was applied to the full standardised embedding matrix (perplexity = 30, PCA initialisation, learning rate set automatically). Strain $\times$ phase centroids were computed in t-SNE space as the mean 2D coordinate over all recordings sharing the same strain, phase, and age group. All quantitative comparisons — cosine similarities, L2 distances, significance tests — were performed directly in the 64-dimensional embedding space without reference to the t-SNE projection.

Genetic Distance

Pairwise genetic distances between strains were computed as a normalised Hamming distance over genome-wide SNP genotypes. For each pair of strains (i, j) , only loci at which both strains carried a homozygous, non-missing genotype were considered (heterozygous calls encoded as H and missing values were excluded). The distance was defined as the fraction of those shared informative loci at which the two strains differed:

$$d(i, j) = \frac{|\{\ell : g_i^\ell \neq g_j^\ell\}|}{|\{\ell : \text{both valid}\}|}.$$

For fully inbred strains this is equivalent to the allele-sharing distance. SNP data were drawn from two panels sourced from the Jackson Laboratory Phenome Database: CGD-MDA1 (~ 470 K SNPs, 33 inbred strains) and CGD-MDA2 (~ 132 K SNPs, 28 BXD strains) [5].

Variable-Length Training

To fully exploit the length-agnostic capability, a variable-length training procedure was implemented, allowing the model to learn from sequences of different durations within a single training run rather than being restricted to a fixed window size.

Pouransari et al. [29] proposed an efficient framework for this purpose, combining two components: dataset decomposition and cyclic learning rate scheduling. Although originally developed for large language model training, the core principles transfer naturally to the present setting.

Dataset decomposition. The dataset decomposition procedure partitions a dataset into buckets of increasing lengths. For a sequence of length l , its binary decomposition $l = 2^{i_1} + 2^{i_2} + \dots + 2^{i_k}$ is computed, and the sequence is split into k adjacent sub-sequences $s_1, \dots, s_k$ of lengths $2^{i_1}, \dots, 2^{i_k}$ respectively, each assigned to the corresponding length bucket D_{i_j} . At training time, given a target batch size of b tokens, exactly $b/2^i$ sequences are sampled from bucket D_i , ensuring a constant token budget per batch regardless of sequence length. The probability of sampling from each bucket can then be defined as a curriculum over training. Pouransari et al. found that a *grow-p2* schedule — in which sampling probability shifts progressively toward longer buckets following powers of two — yields the best performance. This is motivated by the assumption that shorter sequences are easier to learn from and should therefore be presented earlier in training. Sampling without replacement within each bucket ensures full dataset coverage, with sequences of increasing length being encountered as training progresses. The cyclic learning rate schedule complements this by ensuring that all bucket lengths are encountered at comparable learning rates; without it, the model would be trained on long-context sequences only at very small learning rates, limiting their contribution to learning.

Adaptation for hBehaveMAE. Applied directly, this pipeline requires sequences as short as two tokens, which is not compatible with hBehaveMAE’s architecture. Because all sequence lengths must be multiples of the mask unit size, the minimum tractable sequence length was fixed to twice the mask unit length, ensuring that at least one mask unit is masked and one remains visible in the shortest possible training example. Sequence lengths then increase as powers of two from this minimum, following the decomposition scheme of Pouransari et al. Finally, since splitting full behavioural recordings at arbitrary points can introduce incomplete pose observations at sequence boundaries, all sequences were edge-padded on both sides to the nearest multiple of the mask unit size before bucketing.

Additional Results

Reproduction of Results on Shot7M2 and MABe22

Table 1: Baseline reproduction on MABe22. Per-task macro F1 (%) for the hierarchical and non-hierarchical models, comparing our retrained models with the values reported by Stoffl et al. [34] ($\Delta = \text{ours} - \text{paper}$). Strain and lights are sequence-level tasks; the remaining behaviours are frame-level.

Task	Hierarchical			Non-hierarchical		
	Paper	Ours	Δ	Paper	Ours	Δ
Strain	87.14	86.44	−0.70	87.58	86.36	−1.22
Lights	73.43	71.26	−2.17	74.38	71.92	−2.46
Approaches	2.80	2.70	−0.10	3.05	2.66	−0.39
Chases	4.01	3.53	−0.48	4.92	4.41	−0.51
Close	73.57	70.85	−2.72	72.03	68.13	−3.90
Contact	60.55	58.10	−2.45	58.64	54.94	−3.70
Huddles	35.05	34.65	−0.40	33.60	34.82	+1.22
Watching	20.29	19.58	−0.71	17.19	16.62	−0.57
Mean Δ			−1.22			−1.44

Table 2: Baseline reproduction of the non-hierarchical BehaveMAE on Shot7M2. Per-task best-layer macro F1 (%), compared with Stoffl et al. [34] ($\Delta = \text{ours} - \text{paper}$).

Task	Paper	Ours	Δ
<i>Movemes</i>			
L foot	95.10	96.03	+0.93
R foot	95.90	96.10	+0.20
L hand	86.10	85.14	−0.96
R hand	88.80	89.39	+0.59
L elbow	39.80	36.90	−2.90
R elbow	37.30	36.34	−0.96
L knee	43.80	48.94	+5.14
R knee	43.70	47.83	+4.13
Ball	58.90	59.59	+0.69
Switch	20.90	20.99	+0.09
<i>Mean</i>			+0.69
<i>Actions</i>			
Idle	98.40	98.08	−0.32
Move	98.90	98.76	−0.14
Turn	41.60	40.90	−0.70
Dribble	99.00	98.92	−0.08
Hold	48.40	44.83	−3.57
Shoot	11.60	12.67	+1.07
Feint	33.40	37.36	+3.96
Spin	12.30	11.43	−0.87
Sprint	75.40	72.35	−3.05
Force	32.00	20.20	−11.80
<i>Mean</i>			−1.55
<i>Activities</i>			
Casual play	57.10	54.07	−3.03
Intense play	76.50	73.50	−3.00
Dribbling training	76.10	75.41	−0.69
Not playing	97.00	96.70	−0.30
<i>Mean</i>			−1.75
Overall mean Δ			−0.87

To establish a reliable starting point for the extensions developed in this thesis, the hierarchical and non-hierarchical BehaveMAE and hBehaveMAE models of Stoffl et al. [34] were retrained from scratch and evaluated under the original linear-probing protocol. Tables 1–3 report the per-task macro F1 scores (%) alongside the published values and their difference ($\Delta = \text{ours} - \text{paper}$). On MABe22 (Table 1), strain and lights are sequence-level tasks and the remaining behaviours are frame-level; the mean deviations are small and negative (−1.22 for the hierarchical and −1.44 for the non-hierarchical model). The larger *relative* gaps on the social-interaction tasks (e.g. chases, approaches) stem from their very low absolute F1 rather than any substantial loss of performance. On Shot7M2, the non-hierarchical model (Table 2) is reproduced to within an overall mean of −0.87, while the hierarchical model (Table 3) preserves the expected coarse-to-fine ordering across encoder levels, with per-level mean deviations within roughly one F1 point. The single largest discrepancy, the action label *Force* (−11.8 in the non-hierarchical model), corresponds to one of the rarest and lowest-scoring labels in the dataset. Taken together, these deviations are consistent with random-seed variation and confirm a faithful reproduction of the published baselines.

Table 3: Baseline reproduction of the hierarchical hBehaveMAE on Shot7M2. Per-task macro F1 (%) at each of the three encoder levels (L0–L2), compared with Stoffl et al. [34] ($\Delta = \text{ours} - \text{paper}$).

Task	Level 0			Level 1			Level 2		
	Paper	Ours	Δ	Paper	Ours	Δ	Paper	Ours	Δ
<i>Movemes</i>									
L foot	94.4	94.72	+0.32	94.3	94.27	−0.03	76.1	79.18	+3.08
R foot	95.5	95.20	−0.30	95.1	94.77	−0.33	76.6	81.03	+4.43
L hand	89.1	89.81	+0.71	86.2	83.97	−2.23	75.9	75.34	−0.56
R hand	91.3	90.35	−0.95	88.3	89.47	+1.17	76.5	76.53	+0.03
L elbow	49.3	49.51	+0.21	40.8	39.84	−0.96	19.4	19.84	+0.44
R elbow	51.7	47.83	−3.87	40.4	39.57	−0.83	18.9	19.61	+0.71
L knee	59.0	58.28	−0.72	50.1	50.08	−0.02	21.2	21.57	+0.37
R knee	57.0	56.67	−0.33	49.8	48.85	−0.95	20.4	20.90	+0.50
Ball	66.7	66.54	−0.16	63.5	62.07	−1.43	32.4	35.79	+3.39
Switch	18.8	18.62	−0.18	20.4	20.24	−0.16	20.6	21.06	+0.46
<i>Mean</i>			−0.53			−0.58			+1.29
<i>Actions</i>									
Idle	90.4	89.66	−0.74	94.0	93.10	−0.90	98.4	98.48	+0.08
Move	93.4	92.93	−0.47	96.3	95.65	−0.65	99.0	98.92	−0.08
Turn	32.5	31.69	−0.81	37.7	37.52	−0.18	45.9	43.48	−2.42
Dribble	98.6	98.29	−0.31	98.9	98.99	+0.09	98.9	98.99	+0.09
Hold	50.2	50.46	+0.26	48.2	45.39	−2.81	40.4	40.75	+0.35
Shoot	15.0	14.22	−0.78	14.5	13.37	−1.13	20.4	20.59	+0.19
Feint	39.3	37.11	−2.19	34.7	34.86	+0.16	36.2	34.27	−1.93
Spin	10.2	10.59	+0.39	12.1	11.21	−0.89	17.8	15.22	−2.58
Sprint	60.0	62.09	+2.09	73.6	70.86	−2.74	81.0	81.70	+0.70
Force	18.0	16.56	−1.44	37.6	40.74	+3.14	47.1	48.79	+1.69
<i>Mean</i>			−0.40			−0.59			−0.39
<i>Activities</i>									
Casual play	49.6	50.30	+0.70	56.7	56.88	+0.18	63.4	63.11	−0.29
Intense play	60.8	63.00	+2.20	76.0	73.19	−2.81	82.1	83.01	+0.91
Dribbling training	69.1	69.49	+0.39	77.9	77.65	−0.25	80.0	80.26	+0.26
Not playing	94.5	94.27	−0.23	97.2	98.34	+1.14	97.8	97.19	−0.61
<i>Mean</i>			+0.77			−0.43			+0.07

Unsupervised Clustering Recovers Behaviourally Meaningful Syllables

To assess the behavioural relevance of the hBehaveMAE embeddings and demonstrate the potential for concrete behavioural analysis, a simple syllable extraction pipeline was implemented. The last-layer embeddings, at a temporal resolution of 10 frames, were clustered using k -means with $k = 50$. Each cluster was characterised by visually inspecting the median keypoint trajectory of its constituent sequences.

A selection of 5 out of 50 clusters illustrates that the resulting syllables capture behaviourally interpretable movement patterns, including apparent directional locomotion, freezing, and speed variation (Fig. 1). Beyond their kinematic content, the clusters show differential usage across experimental conditions. Cluster 0, characterised by greater locomotor activity, is under-represented in the old age group compared to cluster 10, which shows the opposite pattern, consistent with known age-related reductions in spontaneous locomotion. Cluster 23, corresponding to an apparent slight backward displacement, is under-represented during the habituation phase and enriched in the central region of the arena, making it a plausible candidate for an object-interaction syllable. Finally, the

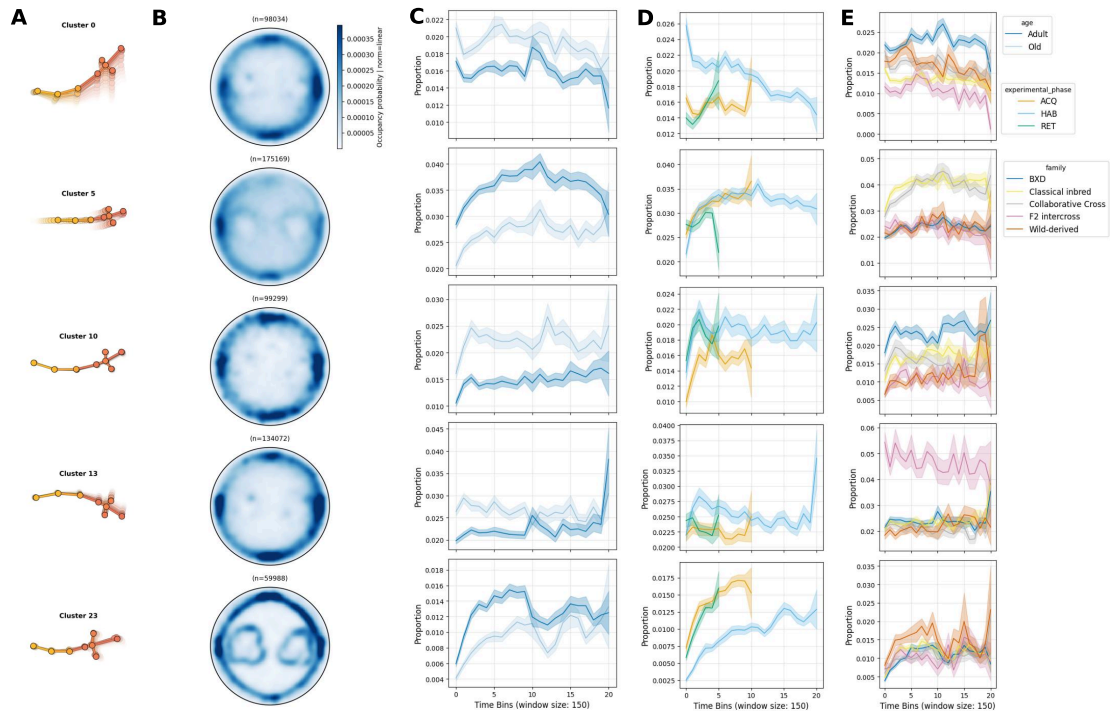


Figure 1: Five out of the 50 clusters obtained by applying K-Means to the hBehaveMAE embeddings were selected as examples of behaviourally relevant clusters. (A) Median pose was computed for each cluster across all elements and plotted with increasing alpha values. (B) The position distribution of each cluster is represented by a heatmap. The difference in cluster expression across time is computed per group of (C) age, (D) experimental phase, and (E) strain family.

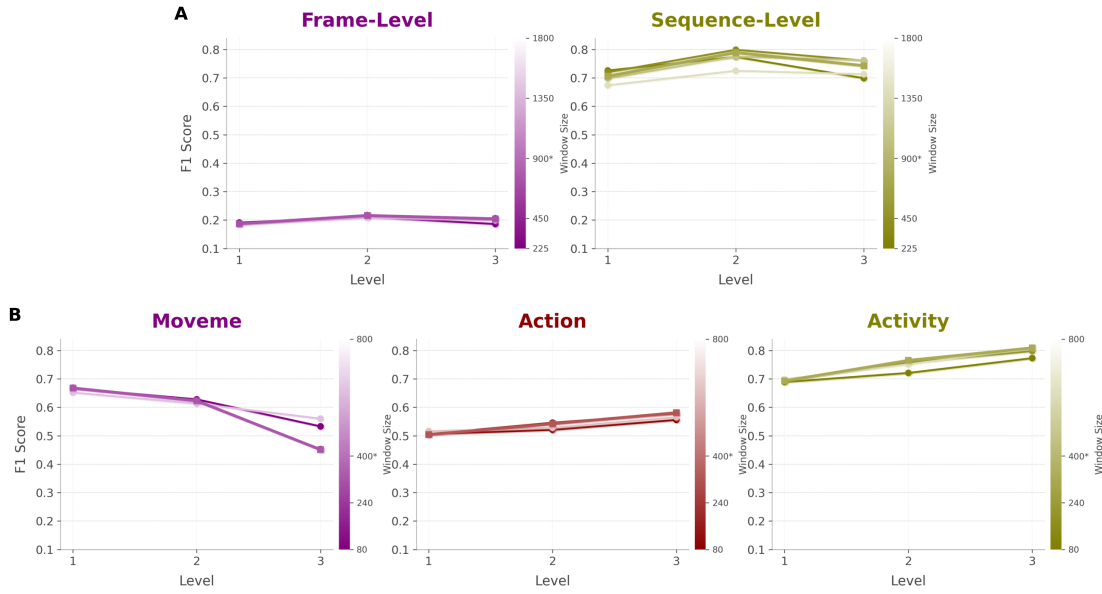


Figure 2: Linear probing of models trained on varying input lengths. The lengths range from the smallest possible window that keeps the exact baseline masking ratio, 225 for MABe22 (A), 80 for Shot7M2 (B), to double the baseline window.

syllable usage profiles also appear to differ across genetic backgrounds, with cluster 13 notably enriched in the F2 intercross family relative to other strain families.

These observations are based on visual inspection and qualitative comparison of syllable usage frequencies; formal statistical testing would be required to establish the significance of these differences. Nevertheless, they demonstrate that the hBehaveMAE embedding space supports the recovery of behaviourally and biologically meaningful structure through simple unsupervised clustering, without any label supervision.

Window Size Study: MABe22 & Shot7M2

With the idea of studying the effect of contextual length, several models were trained with different window lengths, defining the number of frames in the input of the model. Since the window length impacts the dataset partition, the training schedule, the length of attention inside the model, and also the embedding extraction script, significant changes were expected. However, as Figure 2 shows, very little impact is observed on the subsequent linear probing of the embeddings; in particular, no clear correlation is observed between window length and linear probing F1 score.

MABe22 Nested Dropout Linear Probing

Although the baseline hBehaveMAE model did not show promising results on the MABe22 dataset two additional models were trained to study whether nested dropout could improve the hierarchical representation obtained from the embeddings. Figure 3 shows very little impact of nested dropout with on par performance for the hierarchical model and even a significant drop in linear probing performance for the non-hierarchical model. These results were in part expected as the reasons of the poor performance of MABe22 baselines are most probably due to the initial model config used, pooling stages and the defined bottle-neck which nested dropout can not widen.

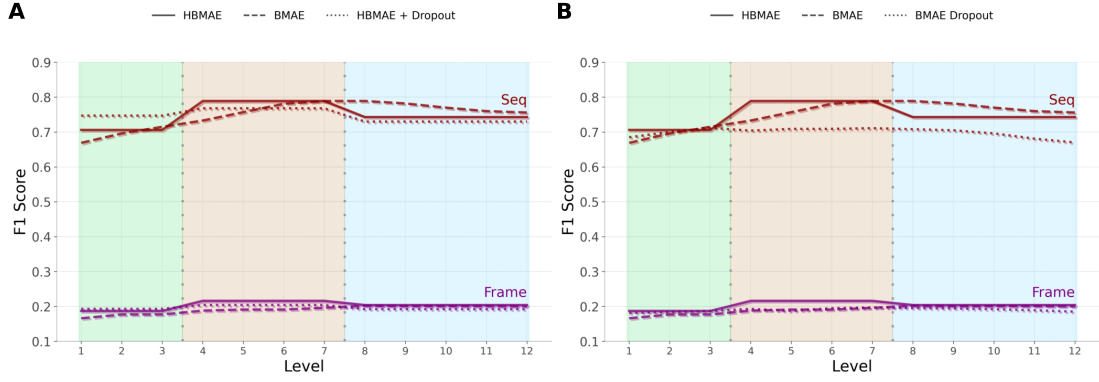


Figure 3: Similar per-frame linear probing was also applied to the MABe22 dataset, here showing the hierarchical model with reverse dropout (A) and the non-hierarchical model with reverse dropout (B), in comparison with both the hierarchical and non-hierarchical baselines.

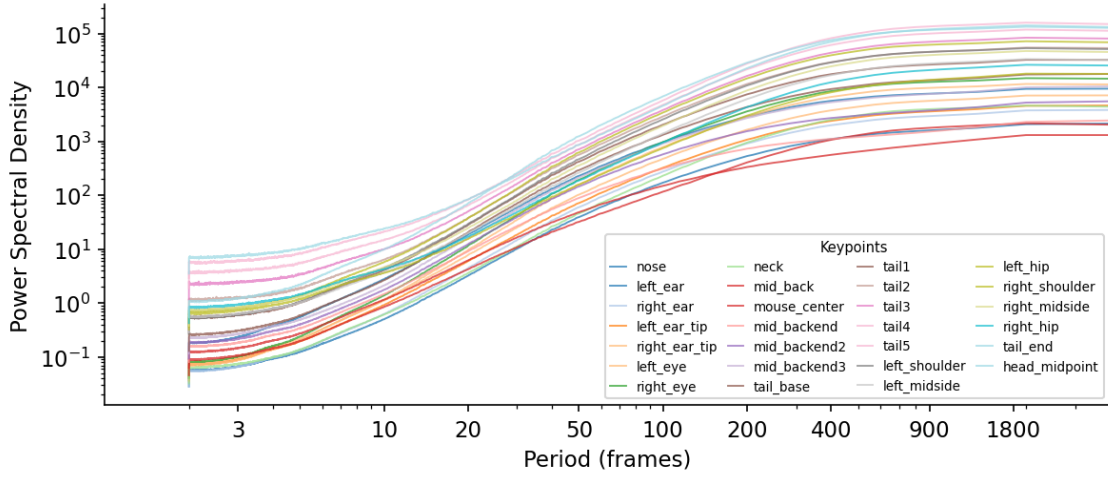


Figure 4: Power spectral density of the Arena keypoint trajectories as a function of period (in frames). The power spectral density of all 27 keypoints is plotted separately and colour-coded.

Frequency Analysis: Arena Dataset

Similar to the MABe22 and Shot7M2 analyses, frequency analysis was computed on the Arena dataset, yielding a smooth power spectral density with no apparent disruptions that could indicate strong temporal patterns in the data (Fig. 4). These results align with those obtained on the MABe22 dataset, reinforcing the claim that the hierarchical structure of natural behaviour is much less apparent than in synthetically generated data. Additionally, the strong correlation between the per-frame linear probing F1 score and the mean explained variance per label attests to the strong correlation between the reconstruction objective and subsequent behavioural analysis (Fig. 5). Interestingly, while our manifold analysis showed a clearer distinction between age groups than experimental groups, the explained variance seems to follow the opposite trend, with lower explained variance for age than for experimental phase.

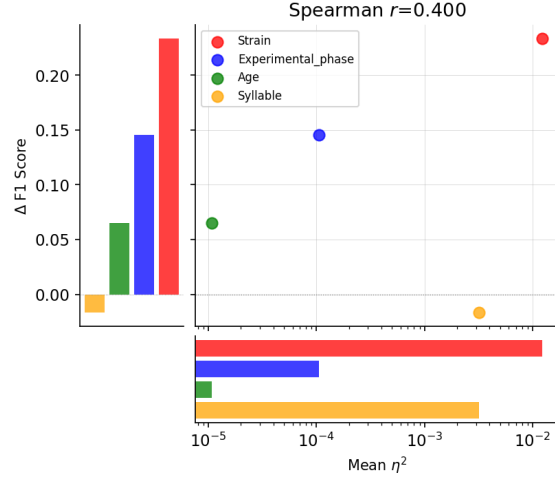


Figure 5: Per-label increase in F1 score between the first and last encoder layers ($\Delta F1$) versus the variance explained by each label (mean η^2) on Arena Dataset, coloured by category, with marginal bar plots. Spearman $r = 0.400$.

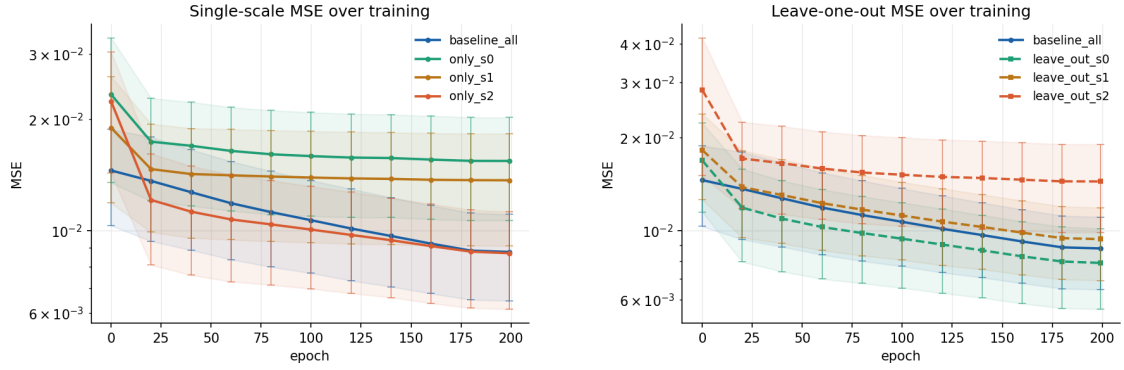


Figure 6: Reconstruction MSE over training epochs for the Arena Dataset length agnostic model. *Left*: single-scale decoding, in which the decoder is given only one level’s embedding (only_s0/s1/s2) or all levels jointly (baseline_all). *Right*: leave-one-out decoding, in which a single level is ablated (leave_out_s0/s1/s2).

Ablation Study - Arena Dataset

The ablation study of the length-agnostic model trained on the Arena dataset shows a strong imbalance in the importance of each layer for reconstruction (Fig. 6). In fact, the last layer is shown to perform on par with, and possibly even better than, the baseline with all fused layers combined. The leave-out results confirm this, with only the leave-out of the last layer performing significantly worse than the baseline. This observed imbalance could be explained by the final temporal resolution (10 frames) being sufficiently low for reconstruction without needing to rely on finer resolutions (1 frame). This could be studied by either increasing the pooling kernels or adding a pooling stage to observe whether the last stage still carries most of the reconstruction task. Finally, this imbalance also strongly advocates for nested dropout, where additional difficulty in the reconstruction task for the last layer could in part resolve this imbalance.

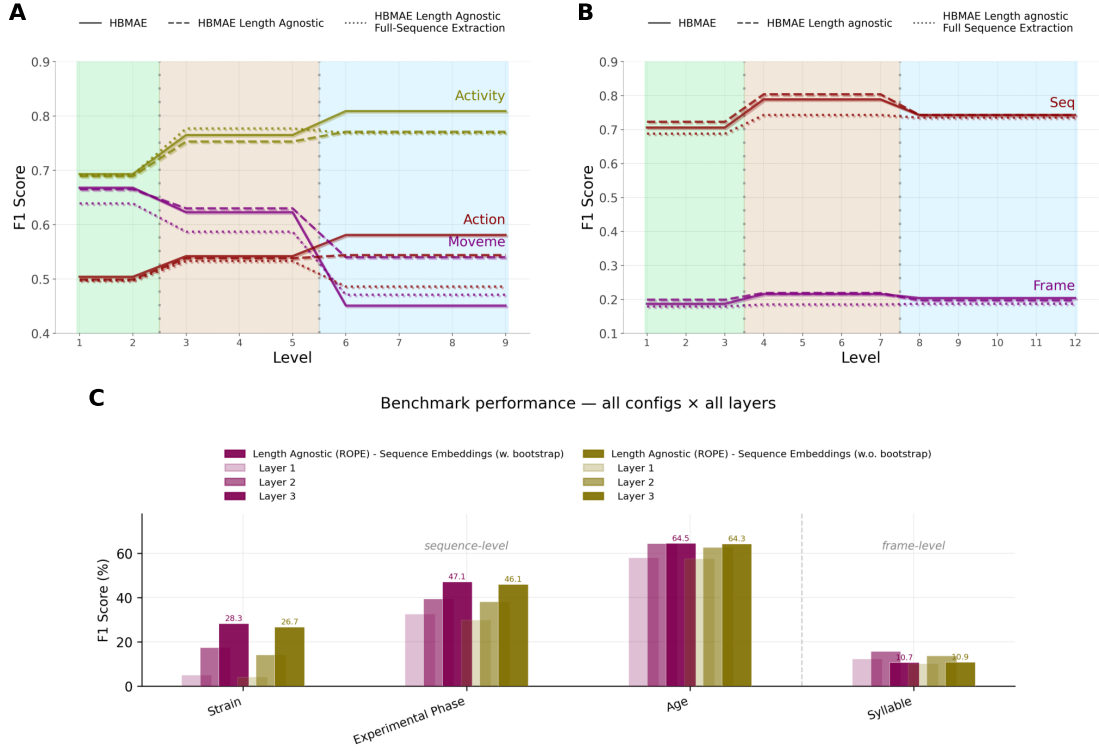


Figure 7: Linear probing of the embeddings obtained from the hierarchical baseline, the length-agnostic model with RoPE temporal positional embeddings, and finally the same model but with full-sequence extracted embeddings, for models trained on Shot7M2 (A) and MABe22 (B). (C) Linear probing comparison of embeddings extracted with the full-sequence method, with and without bootstrap.

Length Agnostic

Although both MABe22 and Shot7M2 are fixed-length datasets, two additional length-agnostic models were trained to study the effect of RoPE positional embeddings. Figure 7A shows on-par linear probing performance for the early layers, with a significant drop for the last layer on Shot7M2. This drop is however not observed on MABe22, where RoPE doesn't seem to hurt performance (Figure 7B). The drop in performance due to full-sequence extraction can be expected, as the maximum length is 1800 frames for both datasets, which is only 4 times the Shot7M2 window length and twice the MABe22 window length. The extra context doesn't compensate for the dense per-frame extraction method in this case. Finally, the impact of bootstrapping in the proposed full-sequence extraction was evaluated on the Arena dataset, where it shows a slight but not drastic improvement (Figure 7C). The F1 score of the last layer's linear probing increases from 26.7 to 28.3, 46.1 to 47.1, and 64.3 to 64.5 for strain, experimental phase, and age respectively.

Variable Length training

The implementation of length-agnostic models showed promising results by allowing the extraction of varying-length embedding representations in a single forward pass. However, all models discussed so far in the study were trained using the script developed by Stoffl et al. [34], using a single fixed window length throughout training. This approach can

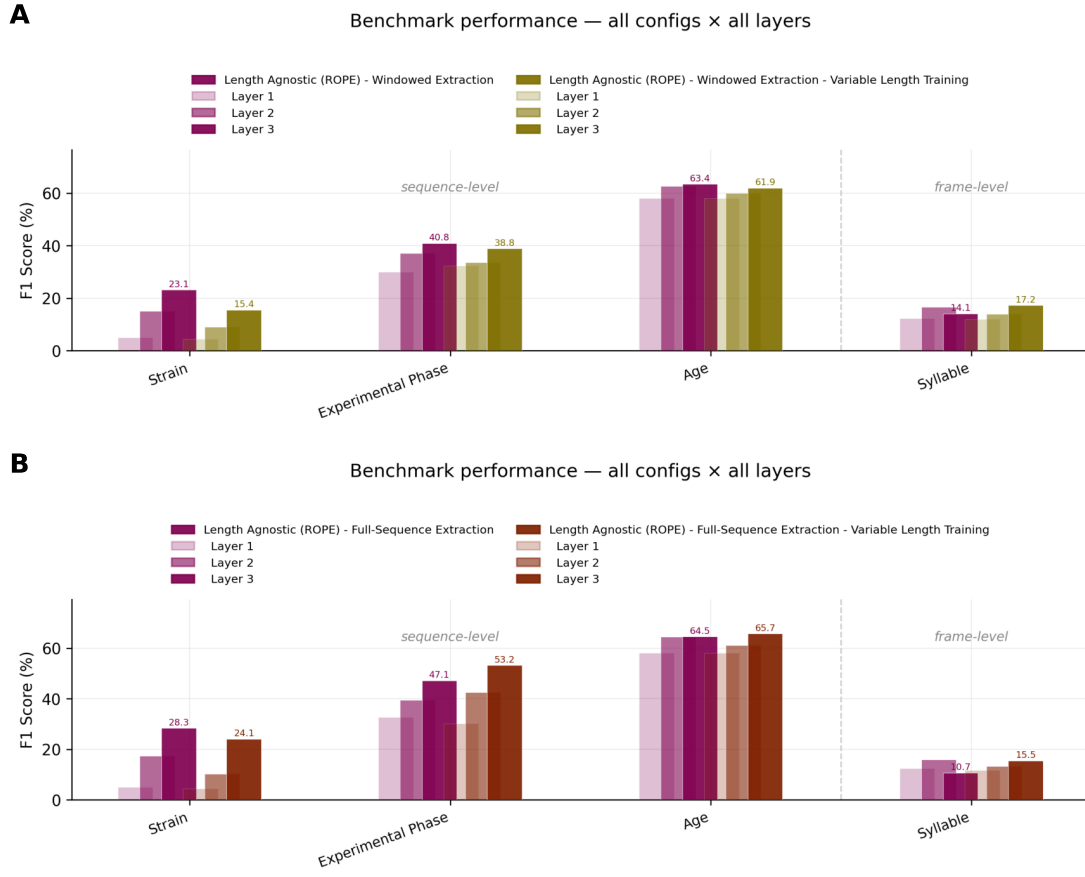


Figure 8: Macro F1 score (%) per label for variable length training versus length agnostic baseline, broken down by encoder level (Layer 1–3, light to dark within each colour). The targets comprise three sequence-level labels (strain, experimental phase, age) and one frame-level label (syllable, from the Keypoint-MoSeq analysis); last layer per-bar values are annotated. Variable length training are evaluated with fixed windowing extraction method (**A**) and varying full-sequence method (**B**), always in comparison to the fixed length training baseline.

be criticised, as the model then receives at extraction time sequences it has never seen during training. Our implementation of distributed dataset and variable-length training was implemented to alleviate this issue and evaluated through per-frame linear probing. Figure 8A shows the linear probing from the windowed extraction, where variable-length training seems to induce a slight decrease in performance, with the strain F1 score decreasing from 23.1 to 15.4. Figure 8B shows that the linear probing of the full-sequence extraction has different effects depending on the predicted label, with a decrease in strain prediction from 28.3 to 24.1 and an increase in experimental phase prediction from 47.1 to 53.2. These results are quite disappointing and advocate for a better implementation of variable-length training, and possibly a simpler one, for example with two training phases: first with a fixed window, and then with full-sequence lengths. In fact, the current strategy leads to varying windows with, at small lengths, significantly different masking ratios, which can negatively impact the training dynamics. Ensuring an equivalent masking ratio in the current implementation could also improve the results.